

Proactive Defense Benchmark against Deepfake Generation

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Abstract

Despite the proliferation of proactive defenses against deepfakes, the lack of a unified evaluation protocol precludes fair comparison and masks critical vulnerabilities. To bridge this gap, we present the first comprehensive benchmark that systematically assesses disruption, robustness, and transferability encompassing pixel, perceptual, and identity metrics. Our extensive analysis reveals that fidelity and identity metrics capture orthogonal performance axes, often leading to conflicting interpretations when relied upon individually. Furthermore, we identify a fundamental trade-off where peak white-box performance signals overfitting, and we introduce a calibrated evaluation to correct generator-induced identity bias. By exposing these blind spots, we establish a rigorous standard to guide the development of genuinely generalizable protections. Project page is available at: <https://proactivedefensebenchmark.github.io/>

1. Introduction

Since the emergence of deepfake generation techniques (Tolosana et al., 2020), publishing facial images online has inherently exposed individuals to the risk of malicious misuse (Harris, 2018; Chesney & Citron, 2019). As deepfake generation becomes increasingly accessible and realistic, post-hoc detection methods (Choi et al., 2024; Ba et al., 2024; Tan et al., 2024; Yang et al., 2025) alone are often insufficient to mitigate harm, as they operate only after manipulated content has already been created. This has motivated increasing interest in proactive defense (Li

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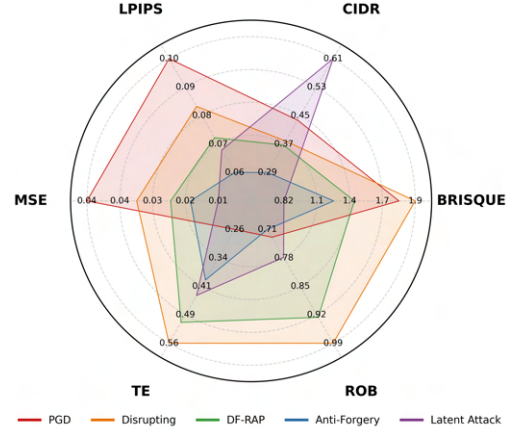


Figure 1. Performance of proactive defenses on SimSwap using multiple evaluation metrics. Additional results for other generators are provided in Appendix B.5.

et al., 2024), where images are protected before publication to interfere with deepfake generation itself.

Recent years have seen a rapid diversification of proactive defense strategies (Ruiz et al., 2020; Zhang et al., 2024; Lee et al., 2025). However, unlike the domain of deepfake detection where unified protocols and comprehensive benchmarks are well-established (Yan et al., 2023; Deng et al., 2024; Le et al., 2025), the evaluation for proactive defense remains critically fragmented. Although recent surveys (Nguyen-Le et al., 2025) exist, they primarily focus on methodological taxonomy rather than providing comprehensive empirical benchmarks. Consequently, existing studies rely on incompatible choices of metrics and generators, rendering cross-paper comparisons ambiguous and difficult. Furthermore, current evaluations frequently overlook the critical gap between ideal settings and real-world deployment. In practice, facial images are frequently subjected to transformations such as JPEG compression and degradation (Pasquini et al., 2021). While these transformations can substantially degrade the effectiveness of proactive defense (Guo et al., 2017; Tian et al., 2024), their effects are often neglected in current protocols. Moreover, assessments of transferability are often misleading. While some prior studies address black-box scenarios, they predominantly rely on a narrow set of target generators or those sharing similar architectures (Wang et al., 2022; Qu et al.,

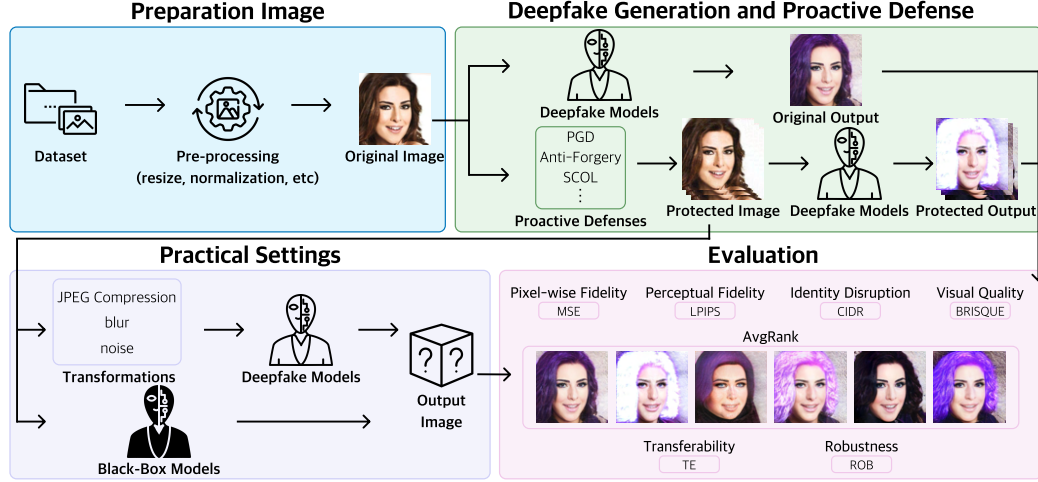


Figure 2. Overview of the proactive defense benchmark considered in this work. The figure illustrates the end-to-end evaluation pipeline, from image preparation and proactive defense to deepfake generation, robustness and transferability analysis, and evaluation.

2024; Shim & Yoon, 2025). This restricted evaluation scope masks the severe performance collapse that typically occurs against architecturally diverse models, thereby risking a significant overestimation of defense generalizability against realistic, unknown threats (Dong et al., 2023).

To address these limitations, establishing a comprehensive benchmark is essential not only for ensuring fair comparison but also for uncovering the complex trade-offs between disruption, robustness, and transferability, as reflected in Figure 1. Through this unified protocol, we aim to shift the research focus from maximizing performance in isolated scenarios to developing defenses that are genuinely robust and generalizable against diverse practical threats.

Our main contributions are summarized as follows:

- We present the first comprehensive benchmark for proactive defense that systematically defines and assesses the complex correlations between the three key factors: **Disruption**, **Robustness**, and **Transferability**. This work provides a rigorous standard for the integrated assessment of diverse defense mechanisms.
- We concretize the definition of “disruption” by establishing a comprehensive taxonomy of evaluation metrics across four key dimensions: **Pixel-level Fidelity**, **Perceptual Fidelity**, **Identity Disruption**, and **Visual Quality**.
- Through extensive empirical experiments, we identify critical blind spots in current practices, revealing a **fundamental trade-off** between optimization specificity and cross-model generalizability. We demonstrate that fidelity and identity metrics capture distinct performance axes, and that peak white-box performance often signals overfitting rather than genuine protection, underscoring the necessity of a balanced evaluation protocol.

2. Related Work

Deepfake Generation Models. Deepfake generation models exhibit significant diversity in their synthesis objectives and architectural designs, which leads to fundamental differences in how facial information is manipulated (Choi et al., 2018; Chen et al., 2020). These approaches can be broadly categorized into attribute manipulation and face swap (Pei et al., 2024). Attribute manipulation models (Choi et al., 2018; Patashnik et al., 2021; Preechakul et al., 2022) aim to modify specific attributes (e.g., hairstyle, age, or facial expression) while preserving the underlying identity. These methods mainly leverage Generative Adversarial Networks (GANs) or Diffusion Models (DMs) to disentangle and traverse the latent representation space for precise or localized control of facial attributes without altering identity. In contrast, face swap models (Chen et al., 2020; Richardson et al., 2021; Shiohara et al., 2023) aim to replace the identity of a target face by injecting identity-related features from a source image while preserving pose, facial expression, and lighting conditions of the target. Earlier face swap approaches (Oldpan, 2018) typically used encoder-decoder structures, in which identity transfer can be achieved through manipulation of latent representations. Subsequent works have incorporated stronger generative priors such as GANs and DMs to improve synthesis fidelity, leading to more diverse methods of deepfake generation (Preechakul et al., 2022; Kim et al., 2022; Zhao et al., 2023; Kim et al., 2025). These continuous architectural evolutions have significantly narrowed the gap between real and generated facial imagery, achieving unprecedented levels of photorealism and identity consistency.

Proactive Defense against Deepfake Generation. Proactive defense aims to disrupt the deepfake generation process

Table 1. Categorization of evaluation metrics. Metrics are grouped by their measurement goals. Metrics adopted in our main benchmark are highlighted in **bold**, while auxiliary metrics evaluated in the Appendix are underlined.

Measurement Goal	Metric	Proposed	Used in Prior Works
Pixel-wise Fidelity	MSE	-	(Ruiz et al., 2020), (Wang et al., 2022), (Qiao et al., 2024), (Qu et al., 2024), (Tang et al., 2024), (Jeong et al., 2025), (Shim & Yoon, 2025)
	<u>PSNR</u>	-	(Wang et al., 2022), (Aneja et al., 2022), (Van Le et al., 2023), (Qiao et al., 2024), (Qu et al., 2024), (Tang et al., 2024), (Jeong et al., 2025), (Lee et al., 2025), (Wang et al., 2025b)
	<u>SR</u>	(Ruiz et al., 2020)	(Ruiz et al., 2020), (Wang et al., 2022), (Qu et al., 2024)
	<u>MAE</u>	-	(Ruiz et al., 2020), (Qu et al., 2024)
	SR_{mask}	(Huang et al., 2022)	(Huang et al., 2022), (Zhang et al., 2024), (Qiao et al., 2024), (Tang et al., 2024), (Qu et al., 2025)
Perceptual Fidelity	LPIPS	(Zhang et al., 2018)	(Aneja et al., 2022), (Van Le et al., 2023), (Qu et al., 2024), (Shim & Yoon, 2025), (Wang et al., 2025b), (Wang et al., 2025a)
	<u>SSIM</u>	(Wang et al., 2004)	(Wang et al., 2022), (Dong et al., 2023), (Qiao et al., 2024), (Qu et al., 2024), (Tang et al., 2024), (Lee et al., 2025), (Wang et al., 2025b)
	<u>FSIM</u>	(Zhang et al., 2011)	(Dong et al., 2023)
	<u>VGG Loss</u>	(Johnson et al., 2016)	(Aneja et al., 2022)
	<u>FID</u>	(Heusel et al., 2017)	(Huang et al., 2022), (Tang et al., 2024)
	<u>CDSR</u>	(Shim & Yoon, 2025)	(Shim & Yoon, 2025)
Identity Disruption	CIDR	ours	-
	<u>ID Loss</u>	(Deng et al., 2019)	(Van Le et al., 2023), (Qu et al., 2024), (Qu et al., 2025), (Jeong et al., 2025), (Shim & Yoon, 2025), (Wang et al., 2025b), (Lee et al., 2025)
	<u>FN_{acc}</u>	(Zhang et al., 2024)	(Zhang et al., 2024)
	<u>FMR</u>	(Wang et al., 2025a)	(Wang et al., 2025a)
Visual Quality	BRISQUE	(Mittal et al., 2012)	(Dong et al., 2023), (Van Le et al., 2023)
	<u>FDFR</u>	(Van Le et al., 2023)	(Van Le et al., 2023)
	<u>TFHC</u>	(zeusees, 2020)	(Huang et al., 2022), (Tang et al., 2024)
	<u>SER-FQA</u>	(Terhorst et al., 2020)	(Van Le et al., 2023)

by preemptively modifying input images prior to their release to prevent downstream misuse. Early studies on deepfake proactive defense adopted pixel-level adversarial attacks designed to maximize distortion in the output space of deepfake generators with imperceptible perturbations (Ruiz et al., 2020; Aneja et al., 2022; Huang et al., 2022). Subsequent studies extended these manipulations beyond raw pixel space or Lab color space (Wang et al., 2022) to more robust feature representations by considering latent space disruption (Shim & Yoon, 2025) or identity disruption (Lee et al., 2025). In addition, identity disruption methods have also been explored under black-box assumptions, disrupting the synthesis process by perturbing the identity extraction process (Wang et al., 2025b) or obfuscating identity-related semantic embeddings (Wang et al., 2025a) rather than requiring access to the deepfake generator itself.

3. Analysis of Proactive Defense Evaluations

To demonstrate the necessity of a comprehensive benchmark, we formalize the evaluation objectives of proactive defenses and identify the systemic limitations inherent in current evaluation protocols.

3.1. Notation and Definitions

Let $\mathcal{X}$ denote the image space. Given a target deepfake generation model $G : \mathcal{X} \rightarrow \mathcal{X}$, the objective of proactive defense is to find a transformation f that solves:

$$\max_f \mathcal{L}(G(x), G(f(x))) \quad \text{s.t. } d(x, f(x)) \leq \epsilon, \quad (1)$$

where $\mathcal{L} : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}^+$ is a discrepancy function (e.g., MSE, LPIPS, or identity distance) measuring the degradation of the synthesis result, with higher values indicating stronger disruption of deepfake generation. d denotes perceptual constraint (typically an ℓ_p -norm) ensuring the protected image $f(x)$ is visually indistinguishable from the original x . Thus, a proactive defense is defined as a transformation function $f_G : \mathcal{X} \rightarrow \mathcal{X}$ that maps an original image $x \in \mathcal{X}$ to a protected counterpart $x_{\text{adv}} = f_G(x)$. In the context of adversarial perturbations, f_G typically adopts an additive form, $f_G(x) = x + \delta$.

In this study, we consider two categories of deepfake generation models, which differ in inputs and synthesis objectives:

- *Attribute Manipulation* aims to modify specific semantic attribute condition c (e.g., age, expression, or hairstyle) of the input image x . The synthesis process, denoted as $G_{\text{attr}}(x, c)$, requires the generator to perform the manipulation while preserving the original identity and image quality.
- *Face Swap* involves a source image x_s (providing identity) and a target image x_t (providing attributes, pose, and illumination). A face swap generator $G_{\text{swap}}(x_s, x_t)$ aims to transfer the source identity onto the target while preserving the target’s structural and visual attributes.

3.2. Defense Objectives and Dimensions

A proactive defense aims to neutralize deepfake generation from the input image x . Given a generation model $G_{\text{att}}(x, c)$, the defense generates a protected image $x_{\text{adv}} =$

$f_{G_{\text{att}}}(\mathbf{x})$ such that the resulting output $G_{\text{att}}(\mathbf{x}_{\text{adv}}, c)$ fails to execute the intended manipulation. In the context of a face swapping model $G_{\text{swap}}(\mathbf{x}_s, \mathbf{x}_t)$, the defense typically produces $\mathbf{x}_{\text{adv}} = f_G(\mathbf{x}_s)$ which induces generation failure, thereby protecting the source image $\mathbf{x}_s$.

The primary challenge in evaluating proactive defenses lies in the lack of standardized success metrics and the disparate assumptions across prior studies. To address this, we formalize evaluation goals by identifying specific synthesis failure modes aligned with the objectives defined in Section 3.1. The effectiveness of these defenses is measured by their ability to induce specific failure modes in deepfake models. We categorize these failures into three primary disruptions:

- **Synthesis Disruption:** A model fails to apply the attribute c or maintain the structural property of the target image.
 - $G_{\text{att}}(\mathbf{x}_{\text{adv}}, c) \neq G_{\text{att}}(\mathbf{x}, c)$
 - $G_{\text{swap}}(\mathbf{x}_{\text{adv}}, \mathbf{x}_t) \neq G_{\text{swap}}(\mathbf{x}_s, \mathbf{x}_t)$
- **Identity Disruption:** The synthesized output fails to preserve the identity of the source image.
 - $\text{id}(\mathbf{x}) \neq \text{id}(G_{\text{att}}(\mathbf{x}_{\text{adv}}, c))$
 - $\text{id}(\mathbf{x}_s) \neq \text{id}(G_{\text{swap}}(\mathbf{x}_{\text{adv}}, \mathbf{x}_t))$
- **Visual Quality Degradation:** The defense induces noticeable artifacts, blurring, or unnatural distortions in the synthesized output.

For notational simplicity in the subsequent analysis, we denote both generative paradigms as G , omit explicit reference to the target attribute c or image $\mathbf{x}_t$, and let $\mathbf{x}$ represent the original image (where $\mathbf{x} = \mathbf{x}_s$ for face swap).

We also characterize proactive defenses across four fundamental technical dimensions to provide a structured analysis of the defensive landscape:

- **Access Regime (White-box vs. Black-box):** We consider white-box settings operating under full access to the generator G , and black-box settings where the defender has no direct knowledge of G and relies on cross-model transferability or surrogate models.
- **Imperceptibility Constraints:** Imperceptibility $d(\mathbf{x}, \mathbf{x}_{\text{adv}})$ can be formalized using pixel-space L_p -norm constraints or perceptual metrics such as LPIPS.
- **Methodologies (Optimization vs. Model-based Learning):** We distinguish between optimization-based approaches that compute an explicit perturbation $\delta = \mathbf{x}_{\text{adv}} - \mathbf{x}$, and model-based approaches that learn a perturbation or reconstruction function f_G to generate protected images.
- **Defense Objectives (disruption vs. identity):** We differentiate between disruption methods aiming for global synthesis failure, and identity-related methods specifically targeting the obfuscation or invalidation of facial identity.

Beyond these core dimensions, additional properties such as ensemble-based strategies, universal perturbations, and robustness-aware training can further differentiate defense

methods. In Appendix B, Table 5 provides a concise overview of representative deepfake generation models and proactive defense methods from prior literature.

4. Our Benchmark

This section introduces our benchmark for the unified evaluation of proactive deepfake defenses. Addressing the limitations identified in Section 3, our benchmark aims to standardize evaluation across three key aspects: (i) diverse deepfake generators and proactive defense methods, (ii) multi-dimensional evaluation metrics that capture distinct synthesis failure behaviors, and (iii) consideration of practical deployment conditions. Figure 2 illustrates our evaluation pipeline. Each defense generates a protected version of an original image, both of which are processed by deepfake generators. We then quantify synthesis disruption, identity disruption, and visual quality by comparing the resulting outputs. To reflect practical deployment scenarios, we further evaluate robustness and transferability.

4.1. Evaluation Datasets and Algorithms

Standard Dataset. We adopt representative face datasets that are well-suited to support both attribute manipulation and face swap tasks. We utilize the CelebA-HQ (Karras et al., 2017) dataset as our primary benchmark due to its high quality and wide adoption in the field. To ensure consistent input dimensions across diverse generators and defense methods, all images are resized to a resolution of 256×256 . Appendix B.1 summarizes datasets commonly used in prior proactive defense studies. To further validate the generalizability of our findings, we additionally conduct experiments on FFHQ and VGGFace2-HQ in Appendix B.3 and B.4, where consistent trends are observed across all metrics and generators.

Deepfake Generators. To evaluate the generation tasks categorized in Section 3.1, we select deepfake generation models: StarGAN (Choi et al., 2018), StyleCLIP (Patashnik et al., 2021), and DiffAE (Preechakul et al., 2022) for attribute manipulation, and SimSwap (Chen et al., 2020), pSp-mix (Richardson et al., 2021), BlendFace (Shiohara et al., 2023), DiffSwap (Zhao et al., 2023), and DiffFace (Kim et al., 2025) for face swap. These models have been widely used in prior proactive defense studies and provide publicly available implementations, facilitating reproducibility and fair comparison.

Proactive Defenses. Regarding proactive defenses, we leverage the taxonomy defined in Section 3.1 to select a diverse set of representative baselines that span various access regimes, methodological paradigms, and defense objectives. As summarized in Table 5(b), our benchmark

includes: PGD (Madry et al., 2017), Disrupting (Ruiz et al., 2020), Anti-Forgery (Wang et al., 2022), DF-RAP (Qu et al., 2024), Latent Attack (Kos et al., 2018), SCOL (Lee et al., 2025), and NullSwap (Wang et al., 2025b). We prioritize official implementations whenever available; for NullSwap, we reproduce the method following the algorithmic details reported in the original work.

4.2. Multi-Dimensional Evaluation Metrics

Designing evaluation metrics for proactive defense poses unique challenges, as effective defenses may induce qualitatively different failure behaviors depending on the target deepfake generation task and model architecture. Guided by the failure mode analysis in Section 3.2, we evaluate the *pro-protectability* of proactive defenses along several complementary dimensions that collectively capture their effectiveness, specificity, and operational reliability. Table 1 categorizes evaluation metrics used in prior work according to their measurement domain and evaluation objective.

- *Synthesis Disruption* can be measured by the discrepancies between $G(\mathbf{x})$ and $G(\mathbf{x}_{\text{adv}})$ using:
 - *Pixel-level Fidelity* captures low-level reconstruction differences such as MSE, $\mathcal{L}_{\text{MSE}}(G(\mathbf{x}), G(\mathbf{x}_{\text{adv}}))$, and PSNR, $\mathcal{S}_{\text{PSNR}}(G(\mathbf{x}), G(\mathbf{x}_{\text{adv}}))$.
 - *Perceptual Fidelity* reflects perceptually meaningful deviations in the generated outputs such as SSIM, $\mathcal{S}_{\text{SSIM}}(G(\mathbf{x}), G(\mathbf{x}_{\text{adv}}))$, and LPIPS, $\mathcal{L}_{\text{LPIPS}}(G(\mathbf{x}), G(\mathbf{x}_{\text{adv}}))$.
- *Identity Disruption* can be evaluated by failures in preserving or transferring identity, measured using a pretrained face recognition model to compute ID Loss ($\mathcal{L}_{\text{ID}}$), face matching rate (FMR), or accuracy (FN_{acc}) (Deng et al., 2019; Wang et al., 2025a; Zhang et al., 2024). In this study, we introduce a calibrated identity ratio (CIDR, detailed below) to decouple the inherent domain shift induced by $G(\cdot)$.
- *Visual Quality Degradation* can be measured by no-reference quality and face-specific operational metrics on $G(\mathbf{x}_{\text{adv}})$, such as BRISQUE, face detection failure rate (FDFR), and true face with high confidence (TFHC).

We provide detailed explanations of all evaluation metrics in Appendix A. While our primary evaluation focuses on the disruption effectiveness, we also provide comparisons between $\mathbf{x}$ and $\mathbf{x}_{\text{adv}}$ in Appendix B.7 to verify the imperceptibility.

Calibrated Identity Metric for Unbiased Evaluation. In most studies, identity disruption is measured by ID Loss using $\mathcal{L}_{\text{ID}}(G(\mathbf{x}), G(\mathbf{x}_{\text{adv}}))$ (Van Le et al., 2023; Shim & Yoon, 2025; Wang et al., 2025b; Lee et al., 2025) or $\mathcal{L}_{\text{ID}}(\mathbf{x}, G(\mathbf{x}_{\text{adv}}))$. However, they often fail to isolate defensive efficacy because ID Loss is inherently biased by the

domain shift and artifacts introduced by the generator G itself. To decouple the defense’s impact from these generative artifacts, we introduce a calibrated identity ratio (CIDR):

$$\mathcal{R}_{\text{ID}}(\mathbf{x}, \mathbf{x}_{\text{adv}}) = 1 - \frac{\mathcal{L}_{\text{ID}}(\mathbf{x}, G(\mathbf{x}))}{\mathcal{L}_{\text{ID}}(\mathbf{x}, G(\mathbf{x}_{\text{adv}}))} \quad (2)$$

By normalizing the distance between the source identity $\mathbf{x}$ and the protected output $G(\mathbf{x}_{\text{adv}})$ against the baseline synthesis error $\mathcal{L}_{\text{ID}}(\mathbf{x}, G(\mathbf{x}))$, CIDR¹ explicitly accounts for the generator’s inherent identity distortion, providing a more objective and unbiased measure of identity obfuscation.

Robustness and Transferability. To evaluate the practical utility of proactive defenses, we consider two critical dimensions beyond basic settings.

- *Robustness to Post-processing Transformations:* Proactive protections often encounter lossy transformations such as JPEG compression and Gaussian blurring before reaching a generator. To quantify defensive resilience under such distortions, we define a relative robustness (ROB) for a given transformation T and metric $\mathcal{L}$ as:

$$\mathcal{R}_{\text{ROB}}(\mathbf{x}, \mathbf{x}_{\text{adv}}; T, \mathcal{L}) = \frac{\mathcal{L}(\mathbf{x}_{\text{ref}}, G(T(\mathbf{x}_{\text{adv}})))}{\mathcal{L}(\mathbf{x}_{\text{ref}}, G(\mathbf{x}_{\text{adv}}))} \quad (3)$$

where $\mathbf{x}_{\text{ref}}$ denotes the reference input (e.g., $\mathbf{x}$ or $G(\mathbf{x})$), or is omitted for no-reference quality measures. This ratio quantifies the relative degradation in defensive effectiveness induced by the transformation T . To obtain a holistic measure of robustness, we report an aggregated ROB¹ for each proactive method, computed by averaging $\mathcal{R}_{\text{ROB}}$ over all evaluated transformations T and metrics $\mathcal{L}$.

- *Cross-generator Transferability:* We assess transferability by applying protected images optimized for a source generator to previously unseen target generators. This setting quantifies the extent to which a defense generalizes across different architectures and generative priors. Formally, given a source generator G_s and a target generator G_t , a protected image $\mathbf{x}_{\text{adv}}^s = f_{G_s}(\mathbf{x})$ is evaluated on G_t . To isolate the transfer effect from the target model’s baseline vulnerability, we define a transfer efficiency (TE) as:

$$\mathcal{R}_{\text{TE}}^{s \rightarrow t}(\mathbf{x}, \mathbf{x}_{\text{adv}}^s, \mathbf{x}_{\text{adv}}^t, \mathcal{L}) = \frac{\mathcal{L}(\mathbf{x}_{\text{ref}}, G_t(\mathbf{x}_{\text{adv}}^s))}{\mathcal{L}(\mathbf{x}_{\text{ref}}, G_t(\mathbf{x}_{\text{adv}}^t))} \quad (4)$$

where $\mathbf{x}_{\text{ref}}$ follows the same convention as above, with G replaced by G_t , and $\mathbf{x}_{\text{adv}}^t = f_{G_t}(\mathbf{x})$. This ratio captures the percentage of the maximum possible degradation achieved through transfer. We report the aggregated TE¹ by averaging these scores across all source-target pairs (G_s, G_t) and evaluation metrics $\mathcal{L}$ to provide each defense’s black-box effectiveness.

¹For numerical stability and consistent aggregation, ROB and TE are clipped at an upper bound of 1, while CIDR is clipped at 0.

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Table 2. Comprehensive evaluation of proactive defense methods across attribute manipulation and face swap tasks. We report pixel-wise fidelity, perceptual fidelity, identity disruption, and visual quality, along with Average Rank (AvgRank), robustness (ROB), and transferability (TE). Additional results are provided in Appendix B.2.

Task	Generator	Metrics	Proactive Defenses						
			White-box				Black-box		
			PGD	Disrupting	DF-RAP	Anti-Forgery	Latent Attack	SCOL	NullSwap
Attribute Manipulation	StarGAN	MSE (↑)	<u>1.3715</u>	0.3003	0.2694	1.6180	0.0958	0.0590	0.0054
		LPIPS (↑)	<u>0.6878</u>	0.4872	0.4477	0.6950	0.4368	0.2317	0.0340
		CIDR (↑)	<u>0.3764</u>	0.1384	0.1002	0.4102	0.1712	0.2390	0.2792
		BRISQUE (↑)	<u>36.4792</u>	25.0217	33.9631	29.6181	37.0453	21.8716	18.3381
		AvgRank (↓)	<u>2.00</u>	4.25	4.50	1.75	4.00	5.50	6.00
		ROB (↑)	0.7008	0.8495	0.8027	0.6253	0.7250	0.9883	<u>0.9787</u>
	TE (↑)	0.4440	<u>0.5755</u>	0.5871	0.5457	0.4290	-	-	
	StyleCLIP	MSE (↑)	5.4086	0.3308	0.3917	<u>0.8045</u>	0.2117	0.0481	0.0108
		LPIPS (↑)	0.7042	0.4069	0.5030	0.4493	<u>0.6439</u>	0.3344	0.1046
		CIDR (↑)	0.5344	0.3079	0.3287	0.2809	0.5852	0.3938	<u>0.5676</u>
		BRISQUE (↑)	31.2795	<u>21.7907</u>	13.3239	21.0689	12.2354	6.5536	12.8330
		AvgRank (↓)	1.50	4.25	3.75	4.00	<u>3.50</u>	5.75	5.25
		ROB (↑)	0.7493	0.9506	0.9480	0.7350	0.8448	0.9921	<u>0.9759</u>
	TE (↑)	0.5075	0.6795	<u>0.6563</u>	0.6157	0.5282	-	-	
	DiffAE	MSE (↑)	0.1186	0.0178	0.0186	0.0020	<u>0.0619</u>	0.0546	0.0053
		LPIPS (↑)	<u>0.4070</u>	0.1757	0.1301	0.0315	0.4598	0.1759	0.0353
		CIDR (↑)	0.1183	0.0275	0.0215	0.0105	0.0117	<u>0.3769</u>	0.5028
		BRISQUE (↑)	96.5489	19.5244	33.4323	12.5174	28.0547	<u>38.4324</u>	28.2317
		AvgRank (↓)	1.75	4.75	4.25	7.00	3.50	<u>2.50</u>	4.25
		ROB (↑)	0.5733	0.8821	0.8196	0.9710	0.7868	<u>0.9283</u>	0.9255
	TE (↑)	0.4514	<u>0.5018</u>	0.5359	0.4262	0.4106	-	-	
SimSwap	MSE (↑)	0.0440	<u>0.0320</u>	0.0238	0.0190	0.0123	0.0050	0.0060	
	LPIPS (↑)	0.0963	<u>0.0801</u>	0.0695	0.0577	0.0653	0.0315	0.0358	
	CIDR (↑)	0.4407	0.3820	0.3708	0.2935	0.6139	0.3519	<u>0.5113</u>	
	BRISQUE (↑)	<u>1.7925</u>	1.9301	1.4151	1.2399	0.8187	1.7359	1.0041	
	AvgRank (↓)	1.75	<u>2.25</u>	3.75	5.25	4.25	5.75	5.00	
	ROB (↑)	0.7269	0.9882	0.9244	0.7077	0.7764	<u>0.9937</u>	0.9981	
TE (↑)	0.4686	0.6866	<u>0.6464</u>	0.5588	0.5166	-	-		
pSp-mix	MSE (↑)	0.5427	0.1270	0.2104	<u>0.2496</u>	0.0830	0.0463	0.0044	
	LPIPS (↑)	0.3331	0.1557	0.1984	<u>0.2038</u>	<u>0.2118</u>	0.1058	0.0254	
	CIDR (↑)	0.1936	0.1090	0.1172	0.0781	<u>0.2069</u>	0.1678	0.2856	
	BRISQUE (↑)	7.1814	<u>6.7925</u>	6.1890	6.2846	5.0783	5.9274	5.6176	
	AvgRank (↓)	1.50	4.25	4.00	<u>3.75</u>	4.00	5.25	5.25	
	ROB (↑)	0.8470	0.9558	0.9476	0.7831	0.8735	<u>0.9981</u>	0.9995	
TE (↑)	0.4954	0.6582	<u>0.6519</u>	0.6049	0.5067	-	-		
BlendFace	MSE (↑)	0.0591	0.0233	0.0088	<u>0.0383</u>	0.0040	0.0013	0.0018	
	LPIPS (↑)	0.0552	0.0291	0.0164	<u>0.0412</u>	0.0151	0.0059	0.0080	
	CIDR (↑)	0.3848	0.2644	0.2077	0.3668	0.5364	0.2293	<u>0.3982</u>	
	BRISQUE (↑)	1.7485	2.1221	2.2879	2.1080	2.3089	<u>2.4045</u>	2.4165	
	AvgRank (↓)	3.00	4.00	4.75	<u>3.50</u>	<u>3.50</u>	5.50	3.75	
	ROB (↑)	0.7876	0.9205	0.9580	0.6122	0.8478	0.9992	<u>0.9924</u>	
TE (↑)	0.4654	0.6362	<u>0.6329</u>	0.5323	0.4645	-	-		
DiffSwap	MSE (↑)	0.0046	0.0046	0.0047	0.0043	0.0097	0.0051	<u>0.0053</u>	
	LPIPS (↑)	0.0293	0.0283	0.0294	0.0309	0.0693	0.0309	<u>0.0316</u>	
	CIDR (↑)	0.0363	0.0425	0.0459	0.0430	0.2674	0.1139	<u>0.2131</u>	
	BRISQUE (↑)	<u>4.0007</u>	3.8500	4.0877	3.5432	3.5636	3.8448	3.7283	
	AvgRank (↓)	5.13	5.38	3.50	5.63	2.25	3.38	<u>2.75</u>	
	ROB (↑)	<u>0.4438</u>	0.4319	0.4594	0.4137	0.3498	0.4395	0.4157	
TE (↑)	0.2895	0.3628	0.4956	0.4128	<u>0.4375</u>	-	-		
DiffFace	MSE (↑)	0.0069	0.0064	0.0068	0.0065	0.0142	0.0101	<u>0.0103</u>	
	LPIPS (↑)	0.0282	0.0269	0.0283	0.0281	0.0528	0.0395	<u>0.0410</u>	
	CIDR (↑)	0.0359	0.0385	0.0534	0.0361	0.6913	0.4120	<u>0.5628</u>	
	BRISQUE (↑)	<u>1.8878</u>	1.7522	1.7187	1.7217	1.3969	2.0021	1.8001	
	AvgRank (↓)	4.50	5.75	4.75	5.75	<u>2.50</u>	<u>2.50</u>	2.25	
	ROB (↑)	0.9968	<u>0.9984</u>	0.9923	1.000	0.8331	0.9807	0.9930	
TE (↑)	0.3015	0.3884	0.4926	0.4316	<u>0.4612</u>	-	-		

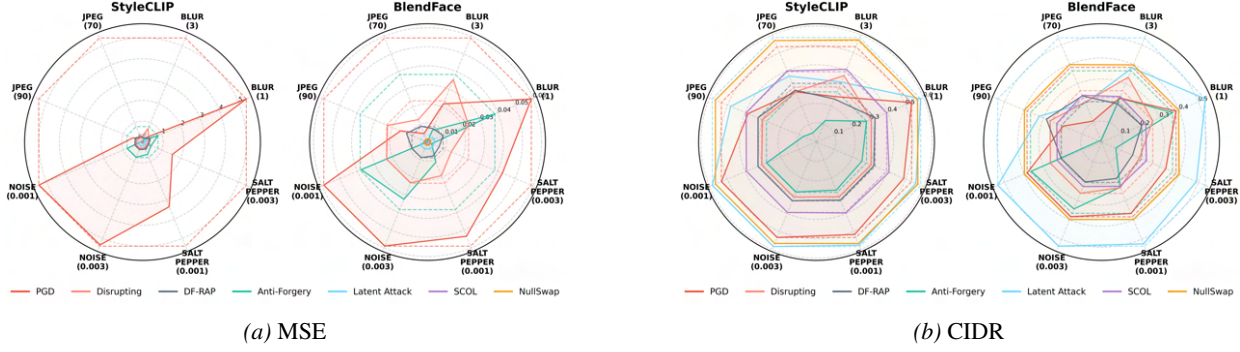


Figure 3. Robustness against post-processing for StyleCLIP and BlendFace. The axes represent transformation types (JPEG compression, blur, Gaussian noise, and salt-and-pepper noise) with varying intensities. Subfigure (a) reports MSE, and subfigure (b) reports CIDR. Solid lines indicate performance under post-processing, while dashed lines denote reference performance without transformations. Complete visualizations for all generators are provided in Appendix C.2.

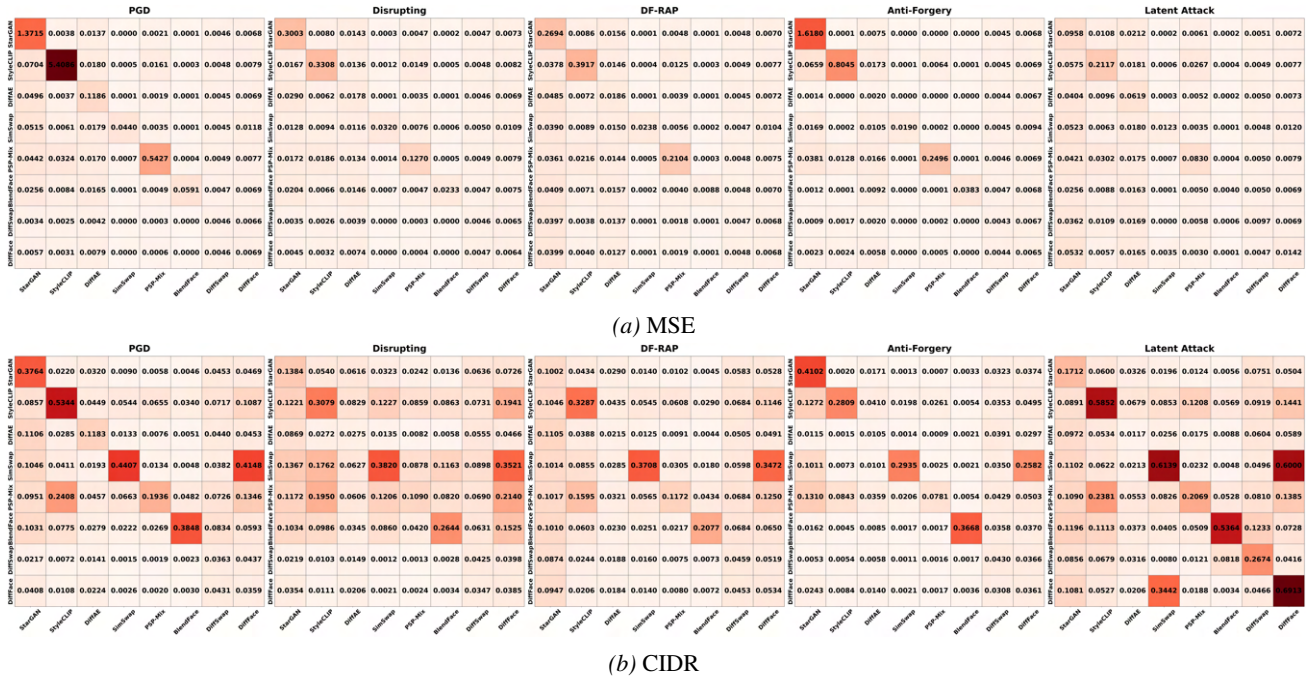


Figure 4. Transferability of proactive defenses across deepfake generators. Rows denote source generators used to generate protected images, and columns denote target generators used for evaluation. Diagonal elements correspond to the white-box setting. Darker colors indicate better performance. (a) reports MSE, and (b) reports CIDR. All heat maps use a uniform color scale for fair comparison. Results for additional metrics are provided in Appendix D.

Unified Performance Aggregation. To assess the holistic effectiveness of proactive defenses, we aggregate performance across data samples and disparate metrics. At the population level, we report dataset-wide averages or ratio-based measures such as the proportion of samples exceeding predefined failure thresholds to ensure operational reliability under diverse inputs. Furthermore, to unify these multi-dimensional measurements into a single interpretable indicator, we employ Average Rank (AvgRank). This approach avoids the pitfalls of directly aggregating metrics with incompatible scales or directions by averaging a de-

fense’s rank across all evaluation criteria. A lower AvgRank indicates that a defense method achieves consistently strong relative performance across multiple generators and metrics.

Correlation of Evaluation Metrics. To mitigate redundancy among the evaluation metrics, we conducted a pairwise correlation analysis across widely adopted metrics (detailed in Appendix E). Our results reveal that metrics sharing similar underlying objectives, such as pixel-wise fidelity (MSE, PSNR) or perceptual fidelity (SSIM, LPIPS), exhibit strong positive correlations. Conversely, metrics

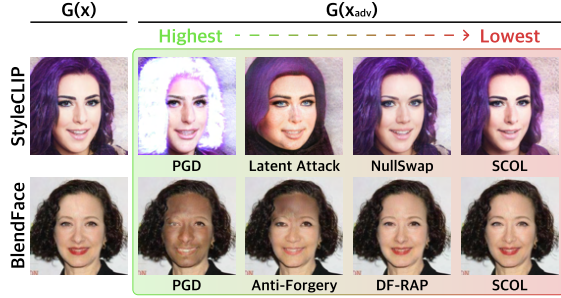


Figure 5. Qualitative comparison of selected proactive defenses based on the AvgRank. We visualize the most effective (Top-1, 2) and the least effective (Top-6, 7) methods for StyleCLIP and BlendFace, according to their AvgRank. The complete rankings of all evaluated methods are provided in Appendix B.6.

across distinct categories show negligible correlation, specifically visual quality and identity disruption. Based on these, we select one representative metric from each category in Table 1 to ensure a concise yet comprehensive evaluation.

5. Experiments and Analysis

Defense performance relies on deepfake generator. Table 2 summarizes the performance of proactive defenses across diverse deepfake generators. Critically, we observe that defense effectiveness varies significantly depending on the target generator. For instance, Anti-Forgery achieves strong disruption on StarGAN, yet it fails to induce meaningful disruption on DiffAE or SimSwap. This indicates that the effectiveness of Anti-Forgery is heavily conditioned on specific generative mechanisms. Conversely, while originally designed strictly for face swap tasks, SCOL demonstrates non-trivial disruption even when extended to unexpected tasks like attribute manipulation. This trend is further amplified in diffusion-based generators such as DiffSwap and DiffFace, where optimization-based methods (e.g., PGD, Disrupting, DF-RAP) show markedly reduced effectiveness despite achieving strong disruption on GAN-based models. This suggests that the iterative denoising process inherent to diffusion models attenuates pixel-level perturbations, making generator architecture a critical factor in defense evaluation. This variability highlights that evaluating defenses on a limited set of generators poses a significant risk of misrepresenting defense capabilities, underscoring the necessity of our comprehensive benchmark.

Improving robustness inevitably degrades disruption.

As illustrated in Figure 3, PGD exhibits the strongest disruption in ideal settings but suffers sharp degradation under post-processing. In contrast, DF-RAP and Disrupting maintain stability against transformations by explicitly incorporating robustness constraints. However, this robustness comes at a cost: these methods exhibit lower visual disruption compared to PGD, confirming that improving

Table 3. Comparative evaluation of ensemble-based defenses against single-source baselines under white-box and black-box settings. Δ denotes the difference between the evaluation scores of ensemble-based defenses and their corresponding single-source baselines.

(a) CMUA vs. Single-source PGD

Model	CMUA		$\Delta_{\text{DiffAE}}^{\text{PGD}}$		$\Delta_{\text{SimSwap}}^{\text{PGD}}$	
	MSE	CIDR	MSE	CIDR	MSE	CIDR
White-box						
DiffAE	0.0211	0.0399	-0.0975	-0.0784	-	-
SimSwap	0.0115	0.3363	-	-	-0.0325	-0.1044
Black-box						
StarGAN	0.0562	0.1117	+0.0066	+0.0011	+0.0047	+0.0071
StyleCLIP	0.0122	0.0798	+0.0085	+0.0513	+0.0061	+0.0387
pSp-mix	0.0062	0.0166	+0.0043	+0.0009	+0.0027	+0.0032
BlendFace	0.0002	0.0073	+0.0001	+0.0022	+0.0001	+0.0025

(b) LEAT vs. Single-source Latent Attack

Model	LEAT		$\Delta_{\text{Lat}}^{\text{DiffAE}}$		$\Delta_{\text{Lat}}^{\text{SimSwap}}$	
	MSE	CIDR	MSE	CIDR	MSE	CIDR
White-box						
DiffAE	0.0605	0.0145	-0.0014	+0.0028	-	-
SimSwap	0.0114	0.5846	-	-	-0.0009	-0.0293
Black-box						
StarGAN	0.0562	0.1117	+0.0009	+0.0080	-0.0110	-0.0050
StyleCLIP	0.0122	0.0798	+0.0003	+0.0231	+0.0036	+0.0143
pSp-mix	0.0062	0.0166	+0.0002	+0.0105	+0.0019	+0.0048
BlendFace	0.0002	0.0103	+0.0000	+0.0015	+0.0001	+0.0055

robustness often compromises white-box performance.

Severe performance drops emerge under unseen generators. Analyzing transferability based on Figure 4 and Table 3, we observe that perturbations optimized for a single source generator exhibit limited transferability to unseen generators. In most cases, defenses that achieve strong disruption on the source generator suffer a sharp decline in effectiveness when transferred to different target generators. This indicates that high white-box scores often result from overfitting rather than genuine protection. In contrast, ensemble-based strategies (CMUA (Huang et al., 2022), LEAT (Shim & Yoon, 2025)) effectively mitigate this drop. Although they exhibit a slight degradation in white-box performance compared to PGD and Latent Attack, they demonstrate improved generalization across diverse architectures. Consequently, the marginal performance drop in ensemble methods is not a limitation but an inevitable trade-off required to ensure cross-model transferability.

Single metrics overlook independent failure modes.

Our empirical results demonstrate that relying on a single metric category leads to conflicting interpretations of defense performance. We observe specific instances where fidelity and identity metrics diverge significantly, confirming that they capture orthogonal dimensions of effectiveness. For instance, NullSwap on DiffAE achieves significant identity disruption (high CIDR) while maintaining near-perfect

	Original Image	G(x)	PGD	Latent Attack	SCOL
DiffAE					
ID Loss		0.4621	0.2988	0.5551	0.3223
CIDR		-	0.1147	0.0000	0.3569

Figure 6. Comparison of identity metrics on DiffAE. In the ID Loss, the value for $G(x)$ denotes $L_{ID}(x, G(x))$, while values for defense methods denote $L_{ID}(G(x), G(x_{adv}))$. CIDR represents the calibrated identity ratio.

pixel fidelity (low MSE). This discrepancy highlights a critical blind spot: relying solely on fidelity metrics would falsely classify strategies with identity disruption objectives as ineffective. To validate the efficacy of unifying these diverse measurements, we examine the qualitative alignment of AvgRank. As visualized in Figure 5, methods with lower AvgRank consistently correspond to stronger qualitative disruption in the generated outputs, confirming that AvgRank effectively summarizes balanced performance across these diverse failure modes.

CIDR corrects identity bias from generators. Our analysis shows that standard ID Loss can provide a distorted view of defense effectiveness due to generator-induced identity bias. As shown in Figure 6, $L_{ID}(x, G(x))$ is already biased, causing relative metrics to produce misleading rankings. For example, Latent Attack obtains a high ID score primarily due to background artifacts, despite largely preserving facial identity, while SCOL induces stronger identity disruption but is ranked lower by conventional metrics. Our proposed CIDR resolves this qualitative-quantitative mismatch by calibrating evaluations to the source identity. After calibration, ineffective methods are correctly suppressed, and effective defenses are re-ranked according to their true identity disruption capability.

Diffusion-based architectures attenuate pixel-level perturbations. We further investigate why optimization-based methods (e.g., PGD, Disrupting, DF-RAP), which maximize pixel-level perturbations via $L(G(x), G(x_{adv}))$, show significantly reduced effectiveness on diffusion-based models (e.g., DiffSwap, DiffFace). To probe this, we vary the number of sampling steps in DiffAE from $T \in \{10, 20, 50, 100, 200\}$ and measure the disruption effectiveness of PGD. As shown in Table 4, increasing the number of sampling steps leads to a consistent degradation in disruption metrics, indicating that the iterative denoising process progressively attenuates pixel-level perturbations. This explains why methods that directly optimize $L(G(x), G(x_{adv}))$ become less effective in diffusion-based architectures. On the other hand, black-box methods such as SCOL and NullSwap maintain disruption performance on these diffusion-based models, despite having no access to the target generator. Since diffusion-based models condition

Table 4. Disruption effectiveness of PGD on DiffAE under varying sampling steps

Metric	T10	T20	T50	T100	T200
MSE $\uparrow$	0.1200	0.0376	0.0224	0.0206	0.0193
LPIPS $\uparrow$	0.4112	0.3988	0.3228	0.3024	0.2791
CIDR $\uparrow$	0.1464	0.0630	0.0360	0.0493	0.0414
BRISQUE $\uparrow$	96.9197	82.1362	65.8337	61.4653	59.4503

their generation process on the source image, disrupting this conditioning stage effectively propagates disruption throughout the entire generation trajectory, regardless of the number of sampling steps.

6. Conclusion

We address the critical fragmentation in proactive defense evaluation by establishing a unified benchmark that integrates disruption, robustness, and transferability. Beyond simply proposing a standardized protocol, our extensive empirical analysis offers a crucial reassessment of the current landscape: we confirm that fidelity and identity metrics operate on orthogonal axes, meaning that singular metrics inevitably yield biased conclusions regarding defense effectiveness. Furthermore, we identify a fundamental trade-off where improving robustness against transformations inherently compromises white-box disruption, suggesting that peak performance in isolated settings often signals overfitting rather than genuine protection. To preclude misleading interpretations caused by generator-induced biases, we demonstrate the necessity of the calibrated identity ratio (CIDR) as a fairer evaluation standard. We hope this unified protocol serves as a foundational guideline for future advancements in securing the digital media landscape.

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Impact Statement

This paper establishes a comprehensive benchmark to rigorously assess the effectiveness of proactive defenses against deepfakes. We do not believe exposing the vulnerabilities of existing defenses will cause harm; rather, it serves as a critical reality check. Many current methods provide a false sense of security, working only in restricted settings while failing against real-world transformations or unknown generators. Since deepfake generation tools are already widely accessible to malicious actors, maintaining the illusion of protection is far more dangerous than revealing their limitations. In the future, as generative models continue to evolve, static defenses may become obsolete. We believe that establishing a standardized and unbiased evaluation protocol provides an essential foundation for the community. By shifting focus from overfitted results to genuine robustness, this work helps prepare society for future threats and guides the development of protection systems that actually work in the wild.

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A. Detailed Evaluation Metrics

We provide comprehensive descriptions of the evaluation metrics used to assess proactive defense methods against deepfake generation 1. While the main manuscript focuses on primary performance indicators, we detail a broader set of metrics for completeness. Most metrics quantify either the discrepancy or similarity between a deepfake output generated from a protected image and a reference baseline, considering two input images, $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{X}$. For notational simplicity, some metrics are formulated as loss functions $\mathcal{L} : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ or similarity functions $\mathcal{S} : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$, where larger values indicate stronger or weaker disruption of deepfake generation, respectively.

Rather than being evaluated on individual image pairs, some metrics provide rate-based or distribution-based measures by using either paired datasets $D = \left\{ \left(\mathbf{x}_1^{(k)}, \mathbf{x}_2^{(k)} \right) \right\}_{k=1, \dots, M}$ consisting of disrupted results and their corresponding reference images or two separate datasets (e.g., a set of disrupted results $D_1 = \left\{ \mathbf{x}_1^{(k)} \right\}_{k=1, \dots, M_1}$ and a set of reference images $D_2 = \left\{ \mathbf{x}_2^{(k)} \right\}_{k=1, \dots, M_2}$). Metrics that require additional information or alternative inputs are described separately below.

A.1. Pixel-wise Fidelity

Mean Absolute Error (MAE) and Mean Squared Error (MSE). MAE and MSE distances quantify the pixel-wise discrepancy between two images. Higher values indicate larger visual distortions caused by the defense, and thus stronger defensive effectiveness.

$$\mathcal{L}_{\text{MAE}}(\mathbf{x}_1, \mathbf{x}_2) = \frac{1}{N} \|\mathbf{x}_1 - \mathbf{x}_2\|_1 \quad (5)$$

$$\mathcal{L}_{\text{MSE}}(\mathbf{x}_1, \mathbf{x}_2) = \frac{1}{N} \|\mathbf{x}_1 - \mathbf{x}_2\|_2^2 \quad (6)$$

where N denotes the total number of pixels across all channels.

Peak Signal-to-Noise Ratio (PSNR). PSNR measures the reconstruction quality between two images at the pixel level. Lower values correspond to more severe degradation of generation quality, indicating stronger defensive impact.

$$\mathcal{S}_{\text{PSNR}}(\mathbf{x}_1, \mathbf{x}_2) = 10 \cdot \log_{10} \left(\frac{\max(\mathbf{x}_1)^2}{\mathcal{L}_{\text{MSE}}(\mathbf{x}_1, \mathbf{x}_2)} \right), \quad (7)$$

where $\max(\mathbf{x}_1)$ denotes the maximum allowable pixel intensity of the image.

Success Rate (SR) (Ruiz et al., 2020). SR measures the proportion of samples for which the defense successfully disrupts the generation process. A sample is considered successful if the $\mathcal{L}_{\text{MSE}}$ distance exceeds a predefined threshold. Higher values indicate that the defense consistently causes substantial deviations in generated outputs.

$$\text{SR}(D) = \frac{1}{M} \sum_{k=1}^M \mathbb{I} \left[\mathcal{L}_{\text{MSE}}(\mathbf{x}_1^{(k)}, \mathbf{x}_2^{(k)}) \geq \tau \right], \quad (8)$$

$\tau = 0.05$ is used in (Ruiz et al., 2020).

Masked Success Rate (SR_{mask}) (Huang et al., 2022). SR_{mask} evaluates defense effectiveness on localized facial regions by restricting the measurement to masked areas where attribute modifications occur. Higher values indicate that the defense consistently causes substantial deviations in generated outputs.

Let $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^{H \times W \times C}$ denote the original and the disrupted images, respectively. We first define a binary spatial mask $\text{Mask}(\mathbf{x}_1, \mathbf{x}_2) \in \{0, 1\}^{H \times W}$ that identifies the region of interest where pixel-wise deviations exceed a threshold τ_m :

$$\text{Mask}_p(\mathbf{x}_1, \mathbf{x}_2) = \mathbb{I} (\|\mathbf{x}_1(p) - \mathbf{x}_2(p)\|_2 > \tau_m), \quad (9)$$

where $p \in \Omega$ denotes the pixel index and $\tau_m = 0.5$ is used in (Huang et al., 2022). The masked MSE distance is then calculated as:

$$\mathcal{L}_{\text{MSE}}^{\text{mask}}(\mathbf{x}_1, \mathbf{x}_2) = \frac{\sum_{p \in \Omega} \text{Mask}_p(\mathbf{x}_1, \mathbf{x}_2) \cdot \|\mathbf{x}_1(p) - \mathbf{x}_2(p)\|_2^2}{\|\text{Mask}(\mathbf{x}_1, \mathbf{x}_2)\|_1}, \quad (10)$$

where $\|\text{Mask}(\mathbf{x}_1, \mathbf{x}_2)\|_1$ denotes the number of active pixels in the mask. Finally, the masked success rate is defined as

$$\text{SR}_{\text{mask}}(D) = \frac{1}{M} \sum_{k=1}^M \mathbb{I} \left[\mathcal{L}_{\text{MSE}}^{\text{mask}}(\mathbf{x}_1^{(k)}, \mathbf{x}_2^{(k)}) > \tau \right] \quad (11)$$

where M is the total number of evaluated image pairs and $\tau = 0.05$ is a predefined success threshold.

A.2. Perceptual Fidelity

Learned Perceptual Image Patch Similarity (LPIPS) (Zhang et al., 2018). LPIPS computes the perceptual distance between two images in the feature space of a pretrained deep network, which correlates well with human visual perception. Higher values indicate larger perceptual differences and more effective disruption by the defense.

$$\mathcal{L}_{\text{LPIPS}}(\mathbf{x}_1, \mathbf{x}_2) = \sum_l \frac{1}{H_l W_l} \sum_{h,w} \|w_l \odot (\Phi_l(\mathbf{x}_1) - \Phi_l(\mathbf{x}_2))\|_2^2 \quad (12)$$

where $\Phi^l : \mathcal{X} \rightarrow \mathbb{R}^{C_l \times H_l \times W_l}$ denotes the feature map extracted from the l -th layer of a pretrained network, H_l and W_l are the spatial height and width of the feature map at layer l , and $w_l \in \mathbb{R}^{C_l}$ is a learned vector that scales each channel based on its perceptual relevance.

Structural Similarity Index (SSIM) (Wang et al., 2004). SSIM evaluates the perceptual similarity between two images by jointly comparing luminance, contrast, and structural information. Lower values indicate greater structural degradation in the generated images, implying a more effective defense.

$$\mathcal{S}_{\text{SSIM}}(\mathbf{x}_1, \mathbf{x}_2) = \frac{(2\mu_{\mathbf{x}_1}\mu_{\mathbf{x}_2} + C_1)(2\sigma_{\mathbf{x}_1\mathbf{x}_2} + C_2)}{(\mu_{\mathbf{x}_1}^2 + \mu_{\mathbf{x}_2}^2 + C_1)(\sigma_{\mathbf{x}_1}^2 + \sigma_{\mathbf{x}_2}^2 + C_2)}, \quad (13)$$

where $\mu_{\mathbf{x}_1}$ and $\mu_{\mathbf{x}_2}$ denote the local mean intensities of $\mathbf{x}_1$ and $\mathbf{x}_2$, respectively; $\sigma_{\mathbf{x}_1}^2$ and $\sigma_{\mathbf{x}_2}^2$ denote the corresponding variances; $\sigma_{\mathbf{x}_1\mathbf{x}_2}$ denotes their covariance; and C_1, C_2 are small constants.

Feature Similarity Index (FSIM) (Zhang et al., 2011). FSIM evaluates image similarity by comparing perceptually important structural features between two images. Lower values indicate stronger disruption of essential structural information.

$$\mathcal{S}_{\text{FSIM}}(\mathbf{x}_1, \mathbf{x}_2) = \frac{\sum_{(i,j) \in \Omega} S_L(\mathbf{x}_1, \mathbf{x}_2; i, j) \cdot PC_m(\mathbf{x}_1, \mathbf{x}_2; i, j)}{\sum_{(i,j) \in \Omega} PC_m(\mathbf{x}_1, \mathbf{x}_2; i, j)} \quad (14)$$

where $S_L(\mathbf{x}_1, \mathbf{x}_2; i, j)$ denotes the local structural similarity at pixel (i, j) computed from phase congruency and gradient magnitude, $PC_m(\mathbf{x}_1, \mathbf{x}_2; i, j)$ represents the corresponding perceptual importance weight, (i, j) index pixel locations, and Ω denotes the image domain.

VGG Loss (Johnson et al., 2016). VGG Loss computes the perceptual and semantic discrepancy between two images using feature representations extracted from a pretrained VGG-19 network. Higher VGG Loss values indicate stronger disruption of high-level semantic and structural information.

$$\mathcal{L}_{\text{VGG}}(\mathbf{x}_1, \mathbf{x}_2) = \sum_l \lambda_l \|\Phi_l(\mathbf{x}_1) - \Phi_l(\mathbf{x}_2)\|_2^2 \quad (15)$$

where $\Phi^l(\mathbf{x})$ denotes the feature map from the l -th layer of the VGG-19 network and λ_l is a hyperparameter that balances the contribution of each layer.

Fréchet Inception Distance (FID) (Heusel et al., 2017). FID measures the distributional distance between two sets of images by comparing the statistics of deep features extracted from a pretrained Inception v3 network. Higher values indicate lower perceptual quality and greater deviation between the two image distributions.

$$\text{FID}(D_1, D_2) = \|\mu_1 - \mu_2\|_2^2 + \text{Tr}(\Sigma_1 + \Sigma_2 - 2(\Sigma_1 \Sigma_2)^{1/2}) \quad (16)$$

where D_1 and D_2 denote two sets of images; (μ_1, Σ_1) and (μ_2, Σ_2) are the mean and covariance of Inception v3 feature embeddings computed over each set; and $\text{Tr}(\cdot)$ denotes the trace operator.

Comprehensive Defense Success Rate (CDSR) (Shim & Yoon, 2025). CDSR provides a holistic evaluation of defense effectiveness by jointly considering pixel-level distortion, perceptual degradation, and identity disruption. A sample is considered successfully defended if *at least one* of the following conditions is satisfied: (i) large pixel-level deviation, (ii) significant identity inconsistency, or (iii) strong perceptual dissimilarity. It is computed as a dataset-level percentage over multiple image pairs. Higher CDSR values indicate that the defense succeeds under at least one of the three complementary criteria, reflecting robust and comprehensive defensive performance.

$$\text{CDSR}(D) = \frac{1}{M} \sum_{k=1}^M \mathbb{I}[(\mathcal{L}_{\text{MSE}}(\mathbf{x}_1^{(k)}, \mathbf{x}_2^{(k)}) > \tau_1) \vee (\mathcal{L}_{\text{ID}}(\mathbf{x}_1^{(k)}, \mathbf{x}_2^{(k)}) > \tau_2) \vee (\mathcal{L}_{\text{LPIPS}}(\mathbf{x}_1^{(k)}, \mathbf{x}_2^{(k)}) > \tau_3)] \quad (17)$$

where $\tau_1 = 0.05$, $\tau_2 = 0.6$, and $\tau_3 = 0.35$ are used in (Shim & Yoon, 2025).

A.3. Identity Disruption

Identity Loss (ID Loss) (Deng et al., 2019). ID Loss measures identity dissimilarity between two images in an embedding space of ArcFace. Higher values indicate greater identity disruption.

$$\mathcal{L}_{\text{ID}}(\mathbf{x}_1, \mathbf{x}_2) = 1 - \frac{\Phi(\mathbf{x}_1)^\top \Phi(\mathbf{x}_2)}{\|\Phi(\mathbf{x}_1)\|_2 \|\Phi(\mathbf{x}_2)\|_2} \quad (18)$$

where $\Phi(\mathbf{x})$ denotes the identity embedding extracted by a pretrained face recognition model.

Face Matching Rate (FMR) (Wang et al., 2025a). FMR measures the proportion of generated images $\mathbf{x}_1$ that are recognized as belonging to the same identity as their corresponding source images $\mathbf{x}_2$ by a face recognition model. Lower FMR values indicate stronger defense.

$$\text{FMR}(D) = \frac{1}{M} \sum_{k=1}^M \mathbb{I}[\|\Phi(\mathbf{x}_1^{(k)}) - \Phi(\mathbf{x}_2^{(k)})\| < \tau] \quad (19)$$

where $\Phi(\cdot)$ denotes the feature extraction by face recognition models and τ is a predefined matching threshold in the face recognition embedding space.

FaceNet Recognition Accuracy Index (FN_{acc}) (Zhang et al., 2024). FN_{acc} measures how effectively a defense prevents FaceNet-based identity recognition systems from classifying face-swapped outputs as the source identity. Lower values indicate stronger defense, as the protected deepfakes are less frequently recognized as belonging to the intended source person. The success rate on identity recognition (SR_{IR}) is defined as an auxiliary measure, using a classifier $c_{FN} : \mathcal{X} \mapsto \mathcal{Y}$ that predicts identity labels from facial embeddings of FaceNet for image $\mathbf{x} \in \mathcal{X}$ (Schroff et al., 2015):

$$\text{SR}_{IR}(D) = \frac{1}{M} \sum_{k=1}^M \mathbb{I}[c_{FN}(\mathbf{x}_{\text{adv}}^{(k)}) = id(\mathbf{x}_{\text{src}}^{(k)})] \quad (20)$$

where $D = \{(\mathbf{x}_{\text{adv}}^{(k)}, \mathbf{x}_{\text{src}}^{(k)})\}$ for the defended images $\mathbf{x}_{\text{adv}}^{(k)}$ and the source images $\mathbf{x}_{\text{src}}^{(k)}$ for faceswap models $id(\cdot)$ is the identity of the source image $\mathbf{x}_2^{(k)}$. FN_{acc} consists of the ratio between SR_{IR} , measuring how often defended face-swapped images in D are recognized as the source identity, compared with the baseline recognition rate for undefended face-swapped images in $D_{\text{undef}} = \{(\mathbf{x}_{\text{undef}}^{(k)}, \mathbf{x}_{\text{src}}^{(k)})\}$.

$$FN_{acc}(D) = \frac{\text{SR}_{IR}(D)}{\text{SR}_{IR}(D_{\text{undef}})} \quad (21)$$

A.4. Visual Quality

BRISQUE (Mittal et al., 2012). BRISQUE is a no-reference image quality assessment metric that evaluates the naturalness and artifact level of generated images without requiring a reference. Higher scores indicate stronger visual artifacts and lower image quality, reflecting a more disruptive defense.

Face Detection Failure Rate (FDFR) (Van Le et al., 2023). FDFR quantifies the proportion of generated images in which no face is detected by a face detection model. Higher values indicate that the defense disrupts facial structure to the extent that generated images are no longer recognized as valid faces.

True Face with High Confidence (TFHC) (zeusees, 2020). TFHC evaluates the proportion of generated faces that are classified as genuine with high confidence by a liveness detection system. Lower values indicate that fewer generated images are recognized as real faces with high confidence, implying that the defense effectively hinders their classification as authentic.

SER-FQA (Terhorst et al., 2020). SER-FQA is a face-specific image quality assessment metric designed to capture subtle degradations in facial structure and perceptual quality. Lower scores indicate significant degradation of facial quality caused by the defense.

B. Deepfake Generation Models and Proactive Methods

Table 5. Overview of deepfake generation models and proactive defense methods. (a) summarizes representative deepfake generation models. (b) overviews proactive defense methods in terms of access regime, methodology, imperceptibility, and publication metadata. (c) summarizes their evaluation coverage across pixel-wise fidelity, perceptual fidelity, identity disruption, visual quality, robustness, and transferability.

(a) Deepfake generation models

Task	Model	Backbone	Venue	Year / Citation	Code
Attribute Manipulation	StarGAN (Choi et al., 2018)	GAN	CVPR	2018 / 5357	✓
	AGGAN (Tang et al., 2019)	GAN	IJCNN	2019 / 192	✓
	StyleCLIP (Patashnik et al., 2021)	GAN	ICCV	2021 / 1574	✓
	DiffAE (Preechakul et al., 2022)	Diffusion	CVPR	2022 / 617	✓
	DiffusionCLIP (Kim et al., 2022)	Diffusion	CVPR	2022 / 934	✓
Face Swap	Faceswap-pytorch (Oldpan, 2018)	Auto Encoder	-	-	✓
	SimSwap (Chen et al., 2020)	Auto Encoder	ACM MM	2020 / 490	✓
	pSp-mix (Richardson et al., 2021)	GAN	CVPR	2021 / 1555	✓
	BlendFace (Shiohara et al., 2023)	Auto Encoder	ICCV	2023 / 91	✓
	DiffSwap (Zhao et al., 2023)	Diffusion	CVPR	2023 / 123	✓
	DiffFace (Kim et al., 2025)	Diffusion	PR	2025 / 112	✓

(b) Proactive defense methods: design and access assumptions

Objective	Method	Access regime	Methodology	Imperceptibility	Venue	Year / Citation	Code
Disruption	Disrupting (Ruiz et al., 2020)	White-box	Optimization	L_p -norm	ECCV	2020 / 187	✓
	Anti-Forgery (Wang et al., 2022)	White-box	Optimization	L_p -norm	IJCAI	2022 / 89	✓
	CMUA (Huang et al., 2022)	White-box	Optimization	L_p -norm	AAAI	2022 / 142	✓
	TCA-GAN (Dong et al., 2023)	Black-box	Model-based Learning	L_p -norm	IEEE TIFS	2023 / 95	-
	ID-Guard (Qu et al., 2025)	White-box	Model-based Learning	L_p -norm	TPAMI	2024 / 2	-
	SUA (Qiao et al., 2024)	White-box	Model-based Learning	L_p -norm	IEEE TIFS	2024 / 16	-
	FOUND (Tang et al., 2024)	White-box	Model-based Learning	L_p -norm	ACM TOMM	2024 / 18	-
	Dual Defense (Zhang et al., 2024)	White-box	Model-based Learning	L_p -norm, SSIM	IEEE TIFS	2024 / 38	✓
	DF-RAP (Qu et al., 2024)	White-box	Optimization	L_p -norm	IEEE TIFS	2024 / 35	✓
	LEAT (Shim & Yoon, 2025)	White-box	Optimization	L_p -norm	ESWA	2025 / 4	-
	Faceshield (Jeong et al., 2025)	White-box	Optimization	L_p -norm	ICCV	2025 / 2	✓
Identity	SCOL (Lee et al., 2025)	Black-box	Model-based Learning	L_p -norm, LPIPS	ACM MM	2025 / 0	✓
	NullSwap (Wang et al., 2025b)	Black-box	Model-based Learning	L_p -norm, LPIPS	ICCV	2025 / 6	-
	FaceSwapGuard (Wang et al., 2025a)	Black-box	Optimization	L_p -norm	-	2025 / 2	-

(c) Evaluation coverage

Method	Pixel-wise Fidelity	Perceptual Fidelity	Visual Quality	Identity Disruption	Robustness	Transferability
Disrupting (Ruiz et al., 2020)	✓	-	-	-	✓	-
Anti-Forgery (Wang et al., 2022)	✓	✓	-	-	✓	✓
CMUA (Huang et al., 2022)	✓	-	-	-	-	✓
TCA-GAN (Dong et al., 2023)	-	✓	✓	-	-	✓
ID-Guard (Qu et al., 2025)	✓	-	-	✓	✓	✓
SUA (Qiao et al., 2024)	✓	✓	-	-	-	✓
FOUND (Tang et al., 2024)	✓	-	-	-	-	✓
Dual Defense (Zhang et al., 2024)	✓	✓	-	-	✓	-
DF-RAP (Qu et al., 2024)	✓	✓	-	-	✓	✓
LEAT (Shim & Yoon, 2025)	✓	✓	-	✓	-	✓
Faceshield (Jeong et al., 2025)	✓	-	-	✓	✓	✓
SCOL (Lee et al., 2025)	✓	✓	-	✓	-	✓
NullSwap (Wang et al., 2025b)	✓	✓	-	✓	-	✓
FaceSwapGuard (Wang et al., 2025a)	-	✓	-	✓	✓	✓
Our Benchmark	✓	✓	✓	✓	✓	✓

B.1. Details of Proactive Defenses

In this section, we provide the detailed mathematical formulations and algorithmic procedures for the proactive defense methods used in our experiment.

Optimization-based Approaches. Optimization-based defenses explicitly solve the maximization problem defined in Eq. 1 by directly optimizing an additive perturbation δ . To encompass both standard and robustness-aware scenarios, we

adopt a general formulation incorporating a transformation function $T(\cdot)$:

$$\delta^* = \arg \max_{\delta} \mathcal{L}(G(\mathbf{x}), G(T(\mathbf{x} + \delta))) , \quad \text{s.t. } \|\delta\| \leq \epsilon \quad (22)$$

Here, $T(\cdot)$ represents a post-processing transformation applied to the protected image. In the standard setting (PGD), T is simply the identity function.

However, to enhance robustness, specific methods define T as a differentiable approximation of image transformations:

- **Disrupting Deepfakes** utilizes operations such as Gaussian blur within T , ensuring the defense remains effective even after low-pass filtering.
- **DF-RAP** incorporates a differentiable JPEG approximation into T to learn perturbations robust to compression.
- **Anti-Forgery** transforms the optimization domain from RGB to the LAB color space. It explicitly optimizes perturbations within the chromatic channels (a and b) while preserving the luminance component (L).
- **Latent Attack** diverges from the pixel-level objective. It replaces the generator G in the formulation with the encoder E , maximizing discrepancy within the latent feature space instead of the final output space.

Model-based Learning Approaches. Unlike optimization-based methods that iteratively compute perturbations for each input instance, model-based defenses train a parameterized reconstruction network $r_{\theta} : \mathcal{X} \rightarrow \mathcal{X}$ to instantaneously generate protected images $\mathbf{x}_{\text{adv}} = r_{\theta^*}(\mathbf{x})$.

$$\theta^* = \arg \min_{\theta} \lambda_{\text{recon}} \mathcal{L}_{\text{recon}}(r_{\theta}(\mathbf{x}), \mathbf{x}) - \lambda_{\text{ID}} \mathcal{L}_{\text{ID}}(r_{\theta}(\mathbf{x}), \mathbf{x}) \quad (23)$$

Here, $\mathcal{L}_{\text{recon}}$ combines MSE and LPIPS to ensure the protected image remains visual fidelity to the original.

- **SCOL** induces manifold collapse by manipulating the generator’s latent representations. Instead of relying on external recognition models, it employs an identity inversion strategy within the latent space. Consequently, the output fails to synthesize the intended source identity, effectively neutralizing the manipulation.
- **NullSwap** targets the identity embedding space under a black-box assumption. It employs adaptive loss weighting to shift the source identity vector toward a null or invalid state. This prevents the generator from extracting consistent identity features, resulting in an output where the identity is unidentifiable or collapsed.

Datasets Used in Prior Proactive Defense Studies. Most prior proactive defense methods are evaluated on a limited set of face datasets, with CelebA-HQ being the most commonly adopted benchmark. For fair comparison with existing works, we also conduct our evaluations on CelebA-HQ.

Table 6. Datasets adopted in prior proactive defense studies.

Dataset	Used
CelebA	(Ruiz et al., 2020), (Wang et al., 2022), (Huang et al., 2022), (Qiao et al., 2024), (Tang et al., 2024), (Qu et al., 2024)
CelebA-HQ	(Aneja et al., 2022), (Van Le et al., 2023), (Shim & Yoon, 2025), (Lee et al., 2025), (Wang et al., 2025a), (Jeong et al., 2025), (Wang et al., 2025b)
CelebAMask-HQ	(Qu et al., 2025)
LFW	(Huang et al., 2022), (Qiao et al., 2024), (Qu et al., 2024), (Wang et al., 2025b), (Qu et al., 2025)
VGGFace2	(Van Le et al., 2023), (Zhang et al., 2024)
VGGFace2-HQ	(Aneja et al., 2022), (Jeong et al., 2025), (Lee et al., 2025)
CASIA-WebFace	(Zhang et al., 2024)
FFHQ	(Aneja et al., 2022), (Qu et al., 2024), (Qu et al., 2025)
Film100	(Huang et al., 2022)
FaceScrub	(Dong et al., 2023)

B.2. Additional Quantitative Results of Table 2

This section reports additional quantitative results that complement Table 2 in the main paper. While Table 2 presents a subset of representative metrics for concise comparison, this appendix provides further evaluation results across multiple generators, defenses, and commonly used metrics.

Table 7. Additional quantitative results for proactive defense methods across multiple deepfake generators and evaluation metrics.

Task	Generator	Metrics	Proactive Defenses						
			White-box				Black-box		
			PGD	Disrupting	DF-RAP	Anti-Forgery	Latent Attack	SCOL	NullSwap
Attribute Manipulation	StarGAN	MSE (↑)	1.3715	0.3003	0.2694	1.6180	0.0958	0.0590	0.0054
		PSNR (↓)	4.7026	11.6732	12.2191	4.0173	16.5134	18.6093	29.2641
		SSIM (↓)	0.1283	0.4556	0.4833	0.1156	0.5125	0.5459	0.9410
		LPIPS (↑)	0.6878	0.4872	0.4477	0.6950	0.4368	0.2317	0.0340
		CIDR (↑)	0.3764	0.1384	0.1002	0.4102	0.1712	0.2390	0.2792
		ID Loss (↑)	0.8912	0.7031	0.6700	0.8923	0.5647	0.2878	0.2684
		BRI (↑)	36.4792	25.0217	33.9631	29.6181	37.0453	21.8716	18.3381
		SR (↑)	1.00	1.00	1.00	1.00	0.94	0.60	0.00
		FDFR (↑)	0.57	0.03	0.01	0.55	0.02	0.00	0.00
	StyleCLIP	MSE (↑)	5.4086	0.3308	0.3917	0.8045	0.2117	0.0481	0.0108
		PSNR (↓)	6.9544	13.2088	11.1592	10.5119	14.7425	19.5295	25.9558
		SSIM (↓)	0.3185	0.6071	0.5106	0.5483	0.4450	0.7050	0.8885
		LPIPS (↑)	0.7042	0.4069	0.5030	0.4493	0.6439	0.3344	0.1046
		CIDR (↑)	0.5344	0.3079	0.3287	0.2809	0.5852	0.3938	0.5676
		ID Loss (↑)	0.6428	0.2900	0.3158	0.2787	0.7477	0.2379	0.5358
		BRI (↑)	31.2795	21.7907	13.3239	21.0689	12.2354	6.5536	12.833
		SR (↑)	1.00	1.00	1.00	1.00	0.99	0.40	0.00
		FDFR (↑)	0.09	0.00	0.00	0.00	0.08	0.00	0.00
	DiffAE	MSE (↑)	0.1186	0.0178	0.0186	0.0020	0.0619	0.0546	0.0053
		PSNR (↓)	15.3987	23.9081	23.5901	36.5044	18.2288	19.1150	29.1722
		SSIM (↓)	0.0288	0.5364	0.4468	0.9393	0.4993	0.6475	0.9235
		LPIPS (↑)	0.4070	0.1757	0.1301	0.0315	0.4598	0.1759	0.0353
		CIDR (↑)	0.1183	0.0275	0.0215	0.0105	0.0117	0.3769	0.5028
		ID Loss (↑)	0.2842	0.0915	0.0750	0.0078	0.5059	0.3269	0.4564
		BRI (↑)	96.5489	19.5244	33.4323	12.5174	28.0547	38.4324	28.2317
		SR (↑)	0.99	0.02	0.02	0.00	0.78	0.49	0.00
		FDFR (↑)	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Face Swap	SimSwap	MSE (↑)	0.0440	0.0320	0.0238	0.0190	0.0123	0.0050	0.0060
		PSNR (↓)	19.9665	21.4425	22.6389	23.7911	25.3029	29.2693	28.3813
		SSIM (↓)	0.7870	0.8168	0.8323	0.8594	0.8256	0.9158	0.9016
		LPIPS (↑)	0.0963	0.0801	0.0695	0.0577	0.0653	0.0315	0.0358
		CIDR (↑)	0.4407	0.3820	0.3708	0.2935	0.6139	0.3519	0.5113
		ID Loss (↑)	0.5753	0.4897	0.4377	0.3338	1.0160	0.3487	0.5737
		BRI (↑)	1.7925	1.9301	1.4151	1.2399	0.8187	1.7359	1.0041
		SR (↑)	0.33	0.10	0.03	0.02	0.00	0.00	0.00
		FDFR (↑)	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	pSp-mix	MSE (↑)	0.5427	0.1270	0.2104	0.2496	0.0830	0.0463	0.0044
		PSNR (↓)	8.8960	16.2547	13.2501	12.5052	17.5581	19.7260	29.7747
		SSIM (↓)	0.4774	0.7232	0.6532	0.6500	0.6378	0.7486	0.9411
		LPIPS (↑)	0.3331	0.1557	0.1984	0.2038	0.2118	0.1058	0.0254
		CIDR (↑)	0.1936	0.1090	0.1172	0.0781	0.2069	0.1678	0.2856
		ID Loss (↑)	0.3643	0.1595	0.1786	0.1443	0.3434	0.1574	0.2376
		BRI (↑)	7.1814	6.7925	6.1890	6.2846	5.0783	5.9274	5.6176
		SR (↑)	1.00	0.77	0.99	1.00	0.66	0.37	0.00
		FDFR (↑)	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	BlendFace	MSE (↑)	0.0591	0.0233	0.0088	0.0383	0.0040	0.0013	0.0018
		PSNR (↓)	19.2648	23.8034	27.4301	21.7302	30.1689	35.0762	33.6978
		SSIM (↓)	0.8994	0.9422	0.9647	0.9227	0.9525	0.9816	0.9764
		LPIPS (↑)	0.0552	0.0291	0.0164	0.0412	0.0151	0.0059	0.0080
		CIDR (↑)	0.3848	0.2644	0.2077	0.3668	0.5364	0.2293	0.3982
		ID Loss (↑)	0.6401	0.4343	0.3252	0.5582	0.8814	0.2779	0.4834
		BRI (↑)	1.7485	2.1221	2.2879	2.1080	2.3089	2.4045	2.4165
		SR (↑)	0.54	0.13	0.00	0.21	0.00	0.00	0.00
		FDFR (↑)	0.00	0.00	0.00	0.00	0.00	0.00	0.00

B.3. Additional Quantitative Results using FFHQ

Table 8. Additional results on FFHQ using the same evaluation setup as Table 2.

Task	Generator	Metrics	Proactive Defenses							
			White-box				Black-box			
			PGD	Disrupting	DF-RAP	Anti-Forgery	Latent Attack	SCOL	NullSwap	
Attribute Manipulation	StarGAN	MSE (↑)	<u>1.2800</u>	0.2417	0.2499	1.4402	0.0817	0.0653	0.0053	
		LPIPS (↑)	<u>0.6847</u>	0.4515	0.4058	0.6915	0.3832	0.2528	0.0387	
		CIDR (↑)	<u>0.3498</u>	0.1107	0.0766	0.3768	0.1360	0.2209	0.2328	
		BRISQUE (↑)	<u>34.6629</u>	24.9974	33.3575	28.6003	35.8447	21.9457	18.7795	
		AvgRank (↓)	<u>2.00</u>	4.50	4.25	1.75	4.00	5.50	6.00	
		ROB (↑)	0.7070	0.8612	0.8116	0.6388	0.7406	<u>0.9875</u>	0.9987	
		TE (↑)	0.3120	<u>0.4456</u>	0.4585	0.3373	0.2929	-	-	
		StyleCLIP	MSE (↑)	4.1773	0.2394	0.3327	<u>0.5960</u>	0.1806	0.0540	0.0107
	LPIPS (↑)		0.6647	0.3772	0.4744	0.4515	<u>0.6248</u>	0.3198	0.1048	
	CIDR (↑)		<u>0.5644</u>	0.3450	0.3470	0.3143	0.6227	0.4226	0.5611	
	BRISQUE (↑)		28.0250	<u>18.5784</u>	12.0067	18.3265	12.7946	7.0981	11.3217	
	AvgRank (↓)		1.25	4.25	4.25	4.00	<u>3.00</u>	5.75	5.50	
	ROB (↑)		0.7691	0.9749	0.9528	0.7534	0.8542	0.9950	<u>0.9846</u>	
	TE (↑)		0.3954	0.5563	<u>0.5249</u>	0.4460	0.4100	-	-	
	DiffAE		MSE (↑)	0.1016	0.0134	0.0158	0.0274	<u>0.0597</u>	0.0548	0.0056
		LPIPS (↑)	<u>0.4120</u>	0.1074	0.0965	0.2240	0.4261	0.2064	0.0469	
		CIDR (↑)	0.1795	0.1062	0.0613	0.0558	0.0538	<u>0.3361</u>	0.4175	
		BRISQUE (↑)	96.9813	18.4010	31.2229	18.4908	31.8727	<u>42.2994</u>	19.9031	
		AvgRank (↓)	1.75	5.50	5.00	4.75	3.25	<u>2.75</u>	5.00	
		ROB (↑)	0.6331	0.9708	0.8997	0.8350	0.7938	0.9258	<u>0.9505</u>	
		TE (↑)	0.3243	<u>0.3692</u>	0.4011	0.3027	0.2843	-	-	
		Face Swap	SimSwap	MSE (↑)	0.0369	<u>0.0279</u>	0.0214	0.0169	0.0131	0.0053
	LPIPS (↑)			0.1151	<u>0.0995</u>	0.0892	0.0745	0.0885	0.0422	0.0435
	CIDR (↑)			0.4160	0.3640	0.3388	0.2902	0.5810	0.3163	0.4597
BRISQUE (↑)	<u>4.8177</u>			5.1509	4.3073	3.4305	2.8553	1.5410	1.8451	
AvgRank (↓)	1.75			<u>2.25</u>	3.50	5.00	3.75	6.75	5.00	
ROB (↑)	0.7154			0.9896	0.9069	0.6596	0.7505	<u>0.9967</u>	0.9993	
TE (↑)	0.3565			0.5587	<u>0.5301</u>	0.3728	0.3858	-	-	
pSp-mix	MSE (↑)			0.2789	0.0625	0.1001	<u>0.1186</u>	0.0532	0.0274	0.0037
	LPIPS (↑)		0.3107	0.1445	0.1857	0.1888	<u>0.2180</u>	0.1131	0.0258	
	CIDR (↑)		<u>0.2100</u>	0.1276	0.1301	0.0847	0.2138	0.1801	0.2818	
	BRISQUE (↑)		<u>5.2977</u>	6.0136	3.7429	4.7629	3.9077	3.9905	3.3939	
	AvgRank (↓)		1.75	4.00	4.50	3.75	<u>3.50</u>	5.00	5.50	
	ROB (↑)		0.9420	1.000	0.9991	0.9962	0.9757	<u>0.9994</u>	1.000	
	TE (↑)		0.3916	0.5488	<u>0.5252</u>	0.4215	0.3770	-	-	
	BlendFace		MSE (↑)	0.0644	0.0282	0.0162	<u>0.0349</u>	0.0059	0.0018	0.0024
LPIPS (↑)			0.0666	0.0365	0.0245	<u>0.0419</u>	0.0249	0.0090	0.0116	
CIDR (↑)			<u>0.4044</u>	0.2861	0.2372	0.3168	0.5614	0.2419	0.3848	
BRISQUE (↑)			-6.8507	-6.4102	-6.1881	-6.3018	-5.6363	<u>-6.1336</u>	-6.1690	
AvgRank (↓)			2.75	4.00	4.75	<u>3.50</u>	2.75	6.00	4.25	
ROB (↑)			0.8074	0.9397	0.9543	0.6324	0.8609	0.9997	<u>0.9995</u>	
TE (↑)			0.3733	0.5419	<u>0.5339</u>	0.3803	0.3598	-	-	
DiffSwap			MSE (↑)	0.0051	0.0050	0.0052	<u>0.0063</u>	0.0115	0.0057	0.0059
	LPIPS (↑)		0.0372	0.0372	0.0378	0.0414	0.0939	0.0406	<u>0.0416</u>	
	CIDR (↑)		0.0435	0.0331	0.0446	0.0524	0.2939	0.1322	<u>0.2249</u>	
	BRISQUE (↑)	-4.8245	<u>-4.7708</u>	-4.6004	-5.2541	-5.1350	-5.0286	-5.0176		
	AvgRank (↓)	5.38	5.63	4.00	4.50	2.25	3.75	<u>2.50</u>		
	ROB (↑)	0.9539	0.9894	0.9745	0.8873	0.8501	0.9762	<u>0.9856</u>		
	TE (↑)	0.1861	0.2620	0.3926	0.2480	<u>0.3357</u>	-	-		
	DiffFace	MSE (↑)	0.0103	0.0106	0.0107	0.0103	0.0196	0.0134	<u>0.0151</u>	
LPIPS (↑)		0.0539	0.0547	0.0547	0.0545	0.0887	0.0664	<u>0.0710</u>		
CIDR (↑)		0.0394	0.0428	0.0447	0.0284	0.6259	0.2994	<u>0.4539</u>		
BRISQUE (↑)		-2.4824	-2.4362	<u>-2.2375</u>	-2.7412	-2.2422	-2.2695	-1.9022		
AvgRank (↓)		6.38	5.13	3.88	6.13	1.50	3.25	<u>1.75</u>		
ROB (↑)		<u>0.5242</u>	0.5201	0.4983	0.5493	0.4644	0.5119	0.4730		
TE (↑)		0.1955	0.2626	0.3988	0.2396	<u>0.3566</u>	-	-		

B.4. Additional Quantitative Results using VGGFace2-HQ

Table 9. Additional results on VGGFace2-HQ using the same evaluation setup as Table 2.

Task	Generator	Metrics	Proactive Defenses						
			White-box				Black-box		
			PGD	Disrupting	DF-RAP	Anti-Forgery	Latent Attack	SCOL	NullSwap
Attribute Manipulation	StarGAN	MSE (↑)	1.2887	0.2424	0.2404	1.4225	0.0737	0.0154	0.0080
		LPIPS (↑)	0.6701	0.4395	0.3876	0.6677	0.3691	0.1073	0.0362
		CIDR (↑)	0.3564	0.0895	0.0346	0.3900	0.1140	0.1563	0.2197
		BRISQUE (↑)	33.6946	23.7841	31.5060	28.4784	34.8548	18.3388	17.9882
		AvgRank (↓)	1.75	4.25	4.50	2.00	4.00	5.50	6.00
		ROB (↑)	0.7068	0.8693	0.8255	0.6329	0.7474	0.9770	0.9986
		TE (↑)	0.4821	0.5810	0.5772	0.5481	0.4139	-	-
	StyleCLIP	MSE (↑)	2.7490	0.2223	0.2908	0.4597	0.1876	0.0092	0.0108
		LPIPS (↑)	0.6537	0.3401	0.4407	0.4166	0.5911	0.1941	0.0951
		CIDR (↑)	0.3909	0.1854	0.2028	0.1996	0.4386	0.2193	0.3832
		BRISQUE (↑)	24.1783	20.5644	11.6138	18.8123	12.2644	5.7525	15.0691
		AvgRank (↓)	1.25	4.50	4.25	3.75	3.25	6.00	5.00
		ROB (↑)	0.7647	0.9670	0.9460	0.7600	0.8353	0.9791	0.9859
		TE (↑)	0.5214	0.6701	0.6350	0.5780	0.4889	-	-
	DiffAE	MSE (↑)	0.0962	0.0170	0.0207	0.0051	0.0644	0.0107	0.0067
		LPIPS (↑)	0.3468	0.1686	0.1273	0.0798	0.4443	0.1396	0.0436
		CIDR (↑)	0.0956	0.0973	0.0690	0.0213	0.0207	0.1517	0.3100
		BRISQUE (↑)	88.9888	17.2793	32.5011	11.9436	31.3392	42.4998	18.1879
		AvgRank (↓)	2.00	4.00	3.75	6.75	3.50	3.50	4.50
		ROB (↑)	0.6439	0.9366	0.8663	0.9376	0.7932	0.9093	0.9431
		TE (↑)	0.4594	0.5158	0.5330	0.4356	0.3979	-	-
Face Swap	SimSwap	MSE (↑)	0.0346	0.0241	0.0169	0.0129	0.0100	0.0029	0.0042
		LPIPS (↑)	0.0701	0.0542	0.0476	0.0369	0.0500	0.0163	0.0229
		CIDR (↑)	0.4183	0.3499	0.3321	0.2678	0.5860	0.2491	0.4913
		BRISQUE (↑)	26.6891	26.4644	25.8577	25.6680	24.7518	25.6100	25.7300
		AvgRank (↓)	1.50	2.50	3.50	5.00	4.25	6.75	4.50
		ROB (↑)	0.7612	0.9927	0.9196	0.7135	0.7847	0.9973	0.9993
		TE (↑)	0.4852	0.6857	0.6362	0.5616	0.4991	-	-
	pSp-mix	MSE (↑)	0.3753	0.0863	0.1345	0.1591	0.0642	0.0051	0.0037
		LPIPS (↑)	0.2871	0.1206	0.1524	0.1605	0.1739	0.0244	0.0188
		CIDR (↑)	0.1359	0.0685	0.0741	0.0597	0.1409	0.0877	0.2004
		BRISQUE (↑)	19.6125	18.1347	18.7614	19.0509	16.4725	18.8466	18.3905
		AvgRank (↓)	1.50	5.25	4.00	3.50	4.00	4.75	5.00
		ROB (↑)	0.8398	0.9605	0.9486	0.7850	0.8389	0.9940	0.9999
		TE (↑)	0.5127	0.6635	0.6433	0.5838	0.4741	-	-
	BlendFace	MSE (↑)	0.0751	0.0337	0.0174	0.0414	0.0070	0.0012	0.0019
		LPIPS (↑)	0.0596	0.0319	0.0200	0.0370	0.0227	0.0052	0.0073
		CIDR (↑)	0.4336	0.3112	0.2319	0.3524	0.5820	0.1641	0.3720
		BRISQUE (↑)	18.9811	19.1206	19.0711	19.1808	19.3738	20.0599	19.6909
		AvgRank (↓)	2.75	4.25	5.00	3.00	3.25	5.50	4.25
		ROB (↑)	0.7969	0.9358	0.9512	0.6181	0.8215	0.9997	0.9995
		TE (↑)	0.4850	0.6471	0.6269	0.5418	0.4452	-	-
	DiffSwap	MSE (↑)	0.0061	0.0057	0.0062	0.0060	0.0143	0.0062	0.0066
		LPIPS (↑)	0.0380	0.0378	0.0390	0.0379	0.0923	0.0395	0.0414
		CIDR (↑)	0.0413	0.0391	0.0337	0.0450	0.2418	0.0639	0.1593
		BRISQUE (↑)	15.5201	15.6934	15.5980	15.8499	14.8338	15.5943	15.3361
		AvgRank (↓)	5.50	5.50	4.63	3.75	2.50	3.13	3.00
		ROB (↑)	0.6744	0.6736	0.6735	0.6751	0.6209	0.6733	0.6699
		TE (↑)	0.3139	0.3766	0.4952	0.3992	0.4271	-	-
	DiffFace	MSE (↑)	0.0125	0.0127	0.0129	0.0135	0.0232	0.0152	0.0166
		LPIPS (↑)	0.0470	0.0477	0.0493	0.0498	0.0804	0.0564	0.0608
		CIDR (↑)	0.0397	0.0330	0.0464	0.0381	0.6275	0.2389	0.4437
		BRISQUE (↑)	9.9247	9.6763	9.5813	10.2244	9.5931	9.3581	9.5360
		AvgRank (↓)	5.25	5.50	4.75	3.75	1.75	4.00	3.00
		ROB (↑)	0.7992	0.8029	0.8015	0.7996	0.7304	0.7989	0.8029
		TE (↑)	0.3158	0.3860	0.4999	0.3996	0.4504	-	-

B.5. Visualization of Principal Results

This section presents a comprehensive visualization of proactive defense performance across all evaluated generators and datasets. As shown in Figure 7 and Figure 8, generators are organized into two categories: attribute manipulation models and face swapping models. Each row corresponds to a generator, while columns represent different datasets (CelebA-HQ, FFHQ, and VGGFace2-HQ).

Each radar plot summarizes multiple evaluation metrics, capturing complementary aspects of performance, including pixel-level fidelity (MSE), perceptual similarity (LPIPS), identity disruption (CIDR), visual quality (BRISQUE), robustness (ROB), and transferability (TE). This visualization complements the detailed analysis on SimSwap in Figure 1 by providing a holistic comparison across diverse generators and data distributions.

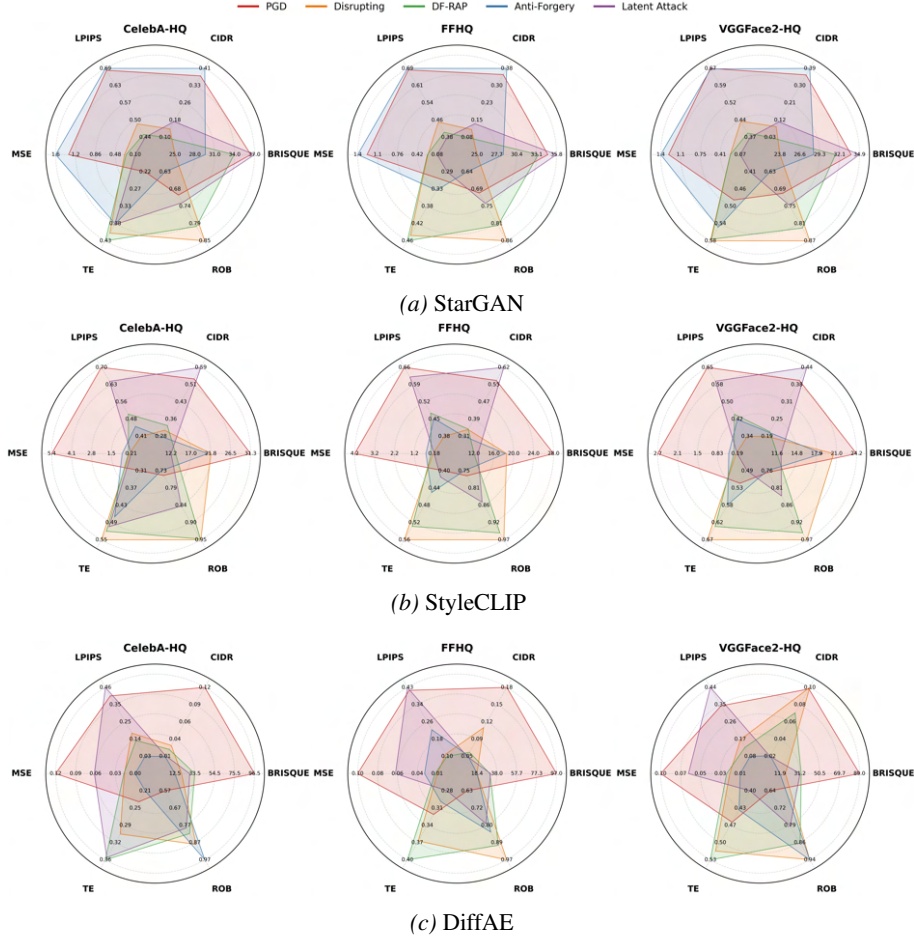


Figure 7. Radar charts of proactive defense performance on attribute manipulation models across three datasets (CelebA-HQ, FFHQ, and VGGFace2-HQ). Rows denote generators and columns denote datasets.

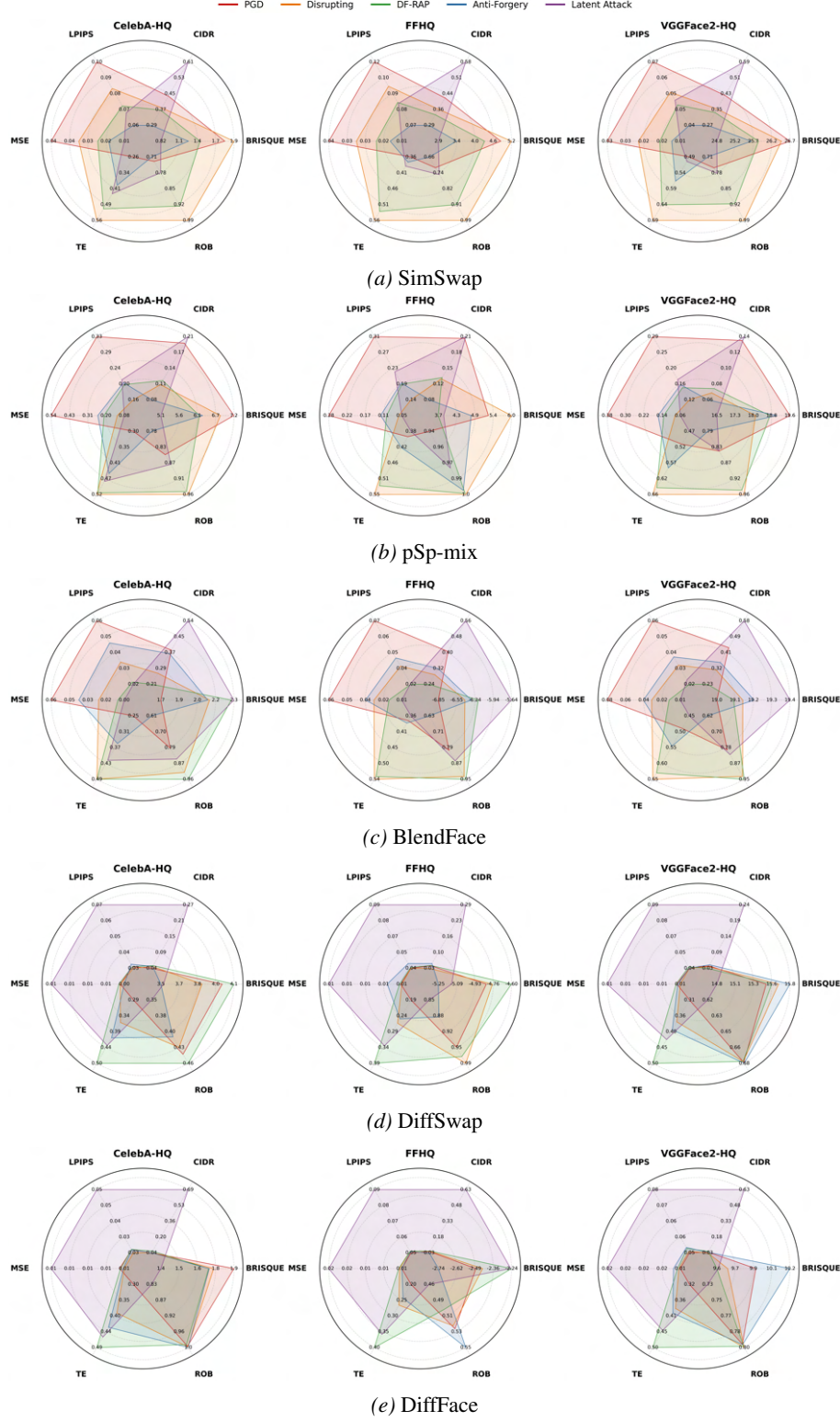


Figure 8. Radar charts of proactive defense performance on face swap models across three datasets (CelebA-HQ, FFHQ, and VGGFace2-HQ). Rows denote generators and columns denote datasets.

B.6. Visualization of AvgRank Results

This section visualizes the AvgRank-based rankings of proactive defenses across all evaluated generators. While Figure 5 presents representative results on a subset of models, we provide complete ranking visualizations here for comprehensive comparison.

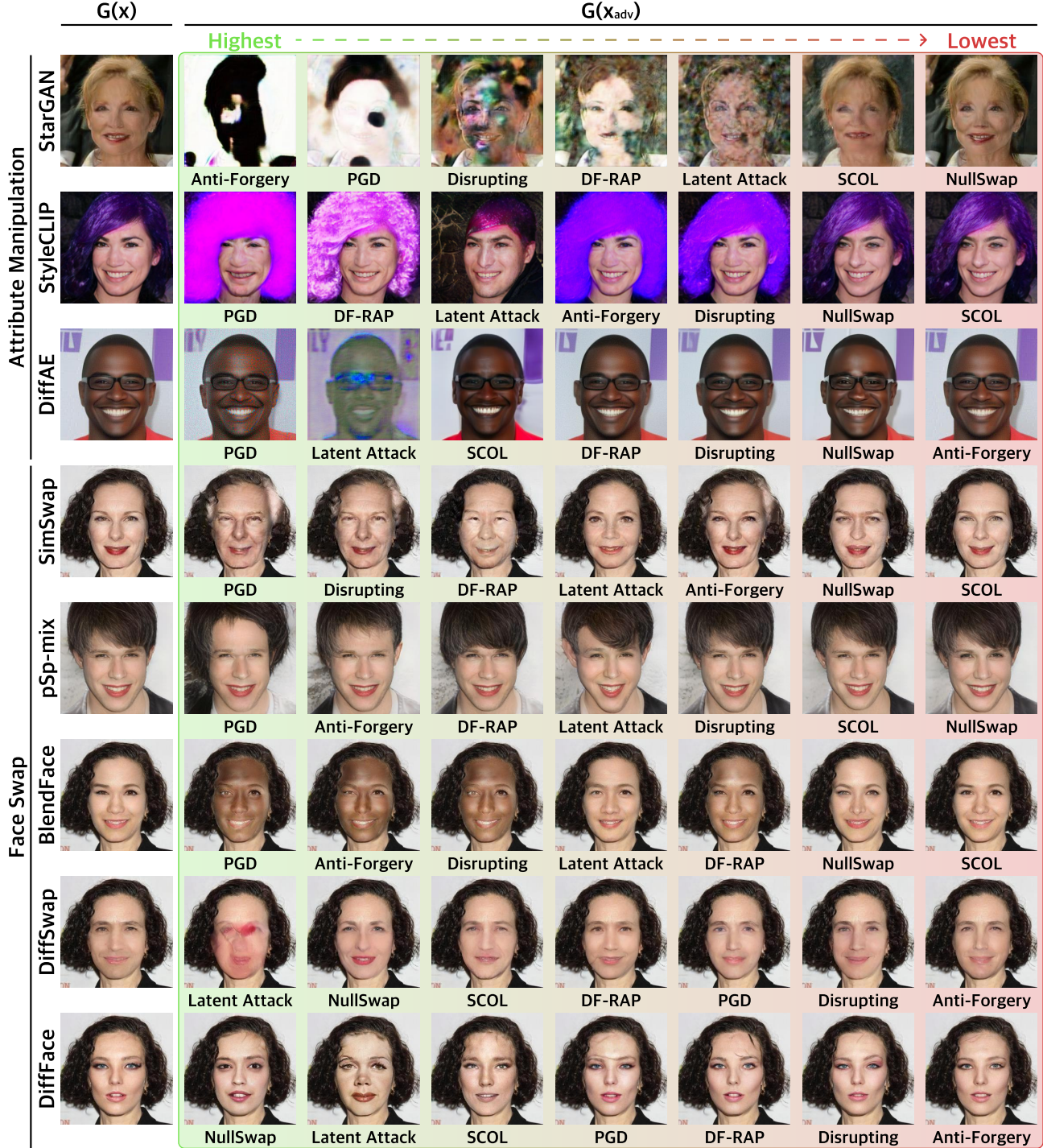


Figure 9. Qualitative comparison of proactive defenses sorted by AvgRank. For each generator, defense methods are arranged in ascending order of their AvgRank (from highest effectiveness on the left to lowest on the right).

B.7. Imperceptibility

To examine the visual imperceptibility of proactive defenses reported in Table 10, we report pixel-wise and perceptual fidelity metrics, including MSE, PSNR, SSIM, and LPIPS, across representative generators and defense methods. Lower MSE and LPIPS values and higher PSNR and SSIM values correspond to higher visual fidelity.

As shown in Figure 10, optimization-based approaches such as DF-RAP and LEAT tend to preserve visual fidelity, as they apply constrained perturbations directly to the original images under a fixed budget. In contrast, model-based learning approaches, including NullSwap and SCOL, tend to exhibit relatively larger visual differences compared to optimization methods, as they rely on image reconstruction mechanisms.

Table 10. The tendency of imperceptibility results for proactive defense methods on StarGAN (attribute manipulation) and SimSwap (face swap). Other generators exhibit similar trends.

Generator	Metric	White-box					Black-box	
		PGD	Disrupting	DF-RAP	Anti-Forgery	LEAT	SCOL	NullSwap
StarGAN	MSE ($\downarrow$)	0.0013	0.0018	0.0012	0.0002	0.0024	0.0617	0.0036
	PSNR ($\uparrow$)	34.7939	33.3090	34.9363	42.4811	32.1176	18.5619	30.5156
	SSIM ($\uparrow$)	0.8705	0.8967	0.8761	0.9764	0.7930	0.5813	0.9456
	LPIPS ($\downarrow$)	0.0225	0.0639	0.0772	0.0004	0.1086	0.2145	0.0137
SimSwap	MSE ($\downarrow$)	0.0009	0.0017	0.0012	0.0004	0.0010	0.0617	0.0036
	PSNR ($\uparrow$)	36.0479	33.6250	35.2073	39.1039	35.9519	18.5619	30.5156
	SSIM ($\uparrow$)	0.8994	0.8951	0.8917	0.9494	0.8977	0.5813	0.9456
	LPIPS ($\downarrow$)	0.0601	0.0641	0.0828	0.0077	0.0621	0.2145	0.0137

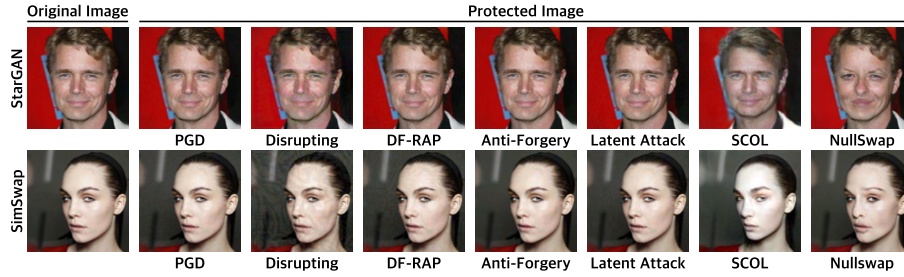


Figure 10. Representative visual comparisons between original images and their protected images produced by different proactive defense methods on StarGAN and SimSwap.

C. Robustness

C.1. post-processing transformations for Robustness

To design a realistic and representative robustness evaluation protocol, we systematically review post-processing perturbations adopted in prior proactive deepfake defense studies. Table 11 summarizes commonly used transformations, including JPEG compression, blur, noise injection, color manipulation, and image reconstruction. Based on this analysis, we select a subset of widely adopted perturbations that reflect practical image degradation scenarios encountered in real-world deployment.

Perturbation Robustness Evaluation	Evaluated in Papers
JPEG Compression	(Wang et al., 2022), (Qu et al., 2024), (Zhang et al., 2024), (Qu et al., 2025), (Jeong et al., 2025), (Wang et al., 2025a)
Gaussian Blur	(Ruiz et al., 2020), (Wang et al., 2022), (Zhang et al., 2024), (Qu et al., 2025), (Wang et al., 2025a)
Box Blur	(Ruiz et al., 2020)
Image Resizing	(Zhang et al., 2024)
Gaussian Noise	(Zhang et al., 2024)
Salt-and-Pepper Noise	(Zhang et al., 2024)
Image Reconstruction (MagDR)	(Wang et al., 2022)

Table 11. Comparison of robustness in proactive deepfake defense methods.

C.2. Visualizations of robustness Results

This section presents comprehensive visualizations of defense robustness across all evaluated generators. While Figure 3 reports representative results on StyleCLIP and BlendFace, we extend the analysis to all generators to provide a complete view of robustness under post-processing transformations.

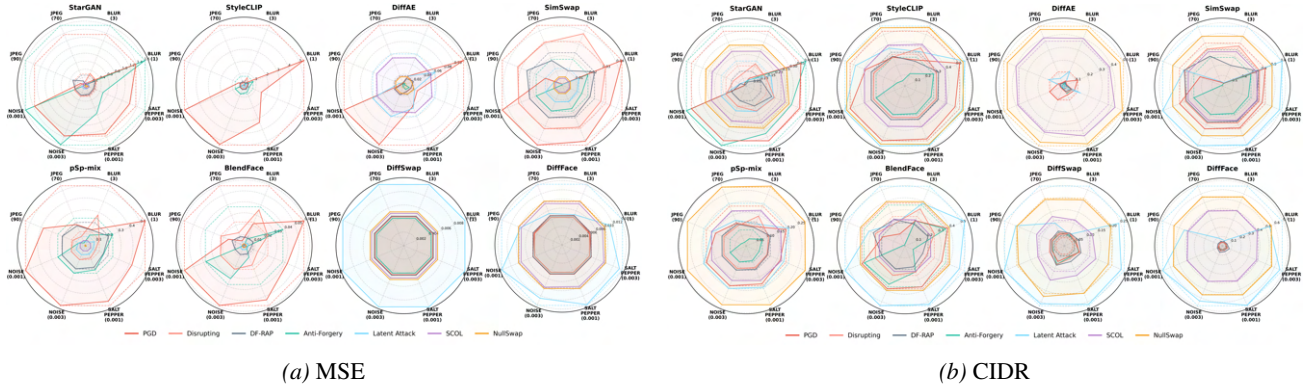


Figure 11. Robustness against post-processing across all evaluated generators. The axes represent transformation types (JPEG compression, Blur, Gaussian Noise, Salt & Pepper) with varying intensities. Solid lines represent performance under transformations, while dashed lines indicate the reference performance without post-processing.

C.3. Quantitative Robustness Results

Following the visual analysis in Appendix C.2, we report detailed quantitative robustness results for each disrupting method in this section. While Figure 11 provides intuitive comparisons under post-processing transformations, the following tables summarize per-metric performance values to support rigorous quantitative evaluation.

Table 12. Quantitative robustness evaluation of PGD under various post-processing transformations.

(a) MSE								
Post-processing	StarGAN	StyleCLIP	DiffAE	SimSwap	pSp-mix	BlendFace	DiffSwap	DiffFace
No Post-processing	1.3715	5.4086	0.1185	0.0440	0.5427	0.0591	0.0046	0.0069
Blur ($\sigma=1$)	1.3715	5.4081	0.1186	0.0440	0.5427	0.0591	0.0046	0.0065
Blur ($\sigma=3$)	0.0562	0.1976	0.0178	0.0007	0.1443	0.0216	0.0046	0.0069
JPEG (Q=70)	0.0021	0.2187	0.0030	0.0010	0.1913	0.0048	0.0046	0.0068
JPEG (Q=90)	0.1190	0.5121	0.0041	0.0123	0.3809	0.0153	0.0046	0.0066
Noise ($\sigma=0.001$)	1.3715	5.3970	0.1184	0.0440	0.5425	0.0591	0.0046	0.0068
Noise ($\sigma=0.003$)	1.3708	5.3416	0.1173	0.0439	0.5419	0.0590	0.0046	0.0069
S&P (p=0.001)	1.3144	3.3594	0.0422	0.0412	0.5037	0.0535	0.0046	0.0069
S&P (p=0.003)	1.2037	1.5547	0.0265	0.0353	0.4232	0.0426	0.0046	0.0068
(b) LPIPS								
Post-processing	StarGAN	StyleCLIP	DiffAE	SimSwap	pSp-mix	BlendFace	DiffSwap	DiffFace
No Post-processing	0.6878	0.7042	0.4070	0.0963	0.3331	0.0552	0.0293	0.0282
Blur ($\sigma=1$)	0.6878	0.7042	0.4070	0.0963	0.3331	0.0552	0.0293	0.0282
Blur ($\sigma=3$)	0.2314	0.4352	0.1854	0.0055	0.1722	0.0261	0.0293	0.0282
JPEG (Q=70)	0.0238	0.4264	0.0325	0.0069	0.1945	0.0091	0.0293	0.0282
JPEG (Q=90)	0.2166	0.5628	0.0529	0.0436	0.2691	0.0199	0.0293	0.0282
Noise ($\sigma=0.001$)	0.6877	0.7041	0.4068	0.0963	0.3330	0.0552	0.0293	0.0282
Noise ($\sigma=0.003$)	0.6876	0.7039	0.4044	0.0963	0.3326	0.0551	0.0293	0.0282
S&P (p=0.001)	0.6793	0.6922	0.2008	0.0929	0.3174	0.0511	0.0293	0.0282
S&P (p=0.003)	0.6693	0.6694	0.1951	0.0860	0.2899	0.0431	0.0293	0.0282
(c) CIDR								
Post-processing	StarGAN	StyleCLIP	DiffAE	SimSwap	pSp-mix	BlendFace	DiffSwap	DiffFace
No Post-processing	0.3765	0.5345	0.1184	0.4407	0.1937	0.3848	0.0363	0.0362
Blur ($\sigma=1$)	0.3765	0.5345	0.1184	0.4407	0.1936	0.3848	0.0477	0.0445
Blur ($\sigma=3$)	0.0531	0.2546	0.0378	0.0284	0.0947	0.2268	0.0527	0.0388
JPEG (Q=70)	0.0213	0.2811	0.0243	0.0394	0.1113	0.1081	0.0524	0.0407
JPEG (Q=90)	0.0105	0.3847	0.0461	0.2289	0.1490	0.1982	0.0546	0.0394
Noise ($\sigma=0.001$)	0.3765	0.5344	0.1183	0.4407	0.1936	0.3847	0.0547	0.0466
Noise ($\sigma=0.003$)	0.3763	0.5339	0.1173	0.4407	0.1934	0.3845	0.0697	0.0514
S&P (p=0.001)	0.3770	0.5174	0.0750	0.4306	0.1845	0.3678	0.0483	0.0515
S&P (p=0.003)	0.3678	0.4904	0.0898	0.4104	0.1656	0.3319	0.0684	0.0501
(d) BRISQUE								
Post-processing	StarGAN	StyleCLIP	DiffAE	SimSwap	pSp-mix	BlendFace	DiffSwap	DiffFace
No Post-processing	36.4792	31.2795	96.5489	1.7925	7.1814	1.7485	4.0007	1.8878
Blur ($\sigma=1$)	36.4822	31.2840	96.5569	1.7925	7.1804	1.7531	-4.8031	1.8878
Blur ($\sigma=3$)	21.4911	33.1753	17.8541	1.2955	7.1325	1.7993	-5.1175	1.8878
JPEG (Q=70)	17.3788	13.0823	16.1227	1.3277	6.3278	2.2660	-4.8531	1.8878
JPEG (Q=90)	21.4017	13.9339	13.7555	1.5447	6.2618	1.8718	-5.0995	1.8878
Noise ($\sigma=0.001$)	36.4790	31.2698	96.4781	1.8038	7.1659	1.7687	-4.9357	1.8876
Noise ($\sigma=0.003$)	36.5107	31.0834	95.8322	1.8021	7.1531	1.7579	-4.4992	1.8878
S&P (p=0.001)	35.8280	26.5648	71.7397	1.6926	7.0919	1.7087	-4.9699	1.8878
S&P (p=0.003)	35.7167	20.1636	39.5108	1.6302	7.0361	1.7185	-4.9134	1.8878

Table 13. Quantitative robustness evaluation of Disrupting under various post-processing transformations.

(a) MSE

Post-processing	StarGAN	StyleCLIP	DiffAE	SimSwap	pSp-mix	BlendFace	DiffSwap	DiffFace
No Post-processing	0.3003	0.3308	0.0178	0.0320	0.1270	0.0233	0.0046	0.0064
Blur ($\sigma=1$)	0.3003	0.3308	0.0178	0.0320	0.1270	0.0233	0.0046	0.0064
Blur ($\sigma=3$)	0.3166	0.6951	0.0160	0.0378	0.2790	0.0354	0.0046	0.0064
JPEG (Q=70)	0.0596	0.2117	0.0044	0.0312	0.0835	0.0127	0.0046	0.0064
JPEG (Q=90)	0.0796	0.2879	0.0066	0.0327	0.1155	0.0228	0.0046	0.0064
Noise ($\sigma=0.001$)	0.3003	0.3305	0.0178	0.0320	0.1269	0.0233	0.0046	0.0064
Noise ($\sigma=0.003$)	0.2998	0.3292	0.0179	0.0320	0.1265	0.0231	0.0046	0.0064
S&P ($p=0.001$)	0.2706	0.2949	0.0133	0.0309	0.1130	0.0190	0.0046	0.0064
S&P ($p=0.003$)	0.2479	0.2433	0.0145	0.0285	0.0918	0.0135	0.0046	0.0064

(b) LPIPS

Post-processing	StarGAN	StyleCLIP	DiffAE	SimSwap	pSp-mix	BlendFace	DiffSwap	DiffFace
No Post-processing	0.4872	0.4069	0.1757	0.0801	0.1557	0.0291	0.0283	0.0269
Blur ($\sigma=1$)	0.4872	0.4069	0.1757	0.0801	0.1557	0.0291	0.0283	0.0269
Blur ($\sigma=3$)	0.3431	0.5391	0.1778	0.0886	0.2418	0.0397	0.0283	0.0269
JPEG (Q=70)	0.1285	0.3555	0.0564	0.0788	0.1283	0.0186	0.0283	0.0269
JPEG (Q=90)	0.1458	0.3962	0.0846	0.0811	0.1490	0.0283	0.0283	0.0269
Noise ($\sigma=0.001$)	0.4872	0.4068	0.1756	0.0800	0.1556	0.0291	0.0283	0.0269
Noise ($\sigma=0.003$)	0.4876	0.4066	0.1747	0.0800	0.1553	0.0289	0.0283	0.0269
S&P ($p=0.001$)	0.5072	0.4079	0.2102	0.0785	0.1510	0.0252	0.0283	0.0269
S&P ($p=0.003$)	0.5354	0.4198	0.2076	0.0757	0.1434	0.0199	0.0283	0.0269

(c) CIDR

Post-processing	StarGAN	StyleCLIP	DiffAE	SimSwap	pSp-mix	BlendFace	DiffSwap	DiffFace
No Post-processing	0.1385	0.3080	0.0275	0.3821	0.1091	0.2645	0.0425	0.0387
Blur ($\sigma=1$)	0.1385	0.3080	0.0276	0.3821	0.1090	0.2645	0.0547	0.0421
Blur ($\sigma=3$)	0.1524	0.3721	0.0454	0.4101	0.1452	0.3336	0.0564	0.0402
JPEG (Q=70)	0.0778	0.2824	0.0295	0.3803	0.0981	0.2051	0.0580	0.0372
JPEG (Q=90)	0.0869	0.3033	0.0434	0.3878	0.1085	0.2658	0.0648	0.0383
Noise ($\sigma=0.001$)	0.1382	0.3079	0.0276	0.3821	0.1090	0.2643	0.0670	0.0521
Noise ($\sigma=0.003$)	0.1403	0.3077	0.0277	0.3821	0.1089	0.2633	0.0828	0.0578
S&P ($p=0.001$)	0.1717	0.3064	0.0568	0.3785	0.1053	0.2353	0.0580	0.0478
S&P ($p=0.003$)	0.1948	0.3066	0.0723	0.3737	0.1013	0.1964	0.0653	0.0600

(d) BRISQUE

Post-processing	StarGAN	StyleCLIP	DiffAE	SimSwap	pSp-mix	BlendFace	DiffSwap	DiffFace
No Post-processing	25.0217	21.7907	19.5244	1.9301	6.7925	2.1221	3.8500	1.7522
Blur ($\sigma=1$)	25.0177	21.7900	19.4999	1.9301	6.7891	2.1205	-5.1296	1.7522
Blur ($\sigma=3$)	23.8909	44.6598	14.1459	1.8704	8.0037	2.1624	-4.9978	1.7522
JPEG (Q=70)	19.6231	19.2952	16.3151	1.8700	6.9016	2.1358	-4.8022	1.7522
JPEG (Q=90)	20.0939	20.8985	15.5738	1.8959	6.7626	2.1550	-4.7816	1.7522
Noise ($\sigma=0.001$)	25.0113	21.7492	19.5536	1.9294	6.7982	2.1143	-4.9020	1.7522
Noise ($\sigma=0.003$)	25.1751	21.5600	19.4216	1.9392	6.7723	2.1359	-4.9118	1.7522
S&P ($p=0.001$)	28.5428	19.7225	51.9769	1.9540	6.6676	2.1561	-4.8912	1.7522
S&P ($p=0.003$)	32.8004	17.7209	56.5645	1.9630	6.6697	2.1447	-4.7649	1.7522

Table 14. Quantitative robustness evaluation of DF-RAP under various post-processing transformations.

(a) MSE

Post-processing	StarGAN	StyleCLIP	DiffAE	SimSwap	pSp-mix	BlendFace	DiffSwap	DiffFace
No Post-processing	0.2694	0.3917	0.0186	0.0238	0.2104	0.0088	0.0047	0.0068
Blur ($\sigma=1$)	0.2694	0.3917	0.0186	0.0238	0.2104	0.0088	0.0047	0.0068
Blur ($\sigma=3$)	0.0185	0.2238	0.0129	0.0106	0.1413	0.0076	0.0047	0.0068
JPEG (Q=70)	0.1142	0.3074	0.0037	0.0181	0.1803	0.0089	0.0047	0.0068
JPEG (Q=90)	0.3342	0.3813	0.0043	0.0260	0.2078	0.0120	0.0047	0.0068
Noise ($\sigma=0.001$)	0.2693	0.3915	0.0186	0.0238	0.2103	0.0088	0.0047	0.0068
Noise ($\sigma=0.003$)	0.2695	0.3907	0.0187	0.0238	0.2101	0.0088	0.0047	0.0068
S&P ($p=0.001$)	0.2417	0.3624	0.0125	0.0227	0.1951	0.0078	0.0047	0.0068
S&P ($p=0.003$)	0.2259	0.3068	0.0139	0.0206	0.1660	0.0065	0.0047	0.0068

(b) LPIPS

Post-processing	StarGAN	StyleCLIP	DiffAE	SimSwap	pSp-mix	BlendFace	DiffSwap	DiffFace
No Post-processing	0.4477	0.5030	0.1301	0.0695	0.1984	0.0164	0.0294	0.0283
Blur ($\sigma=1$)	0.4477	0.5030	0.1301	0.0695	0.1984	0.0164	0.0294	0.0283
Blur ($\sigma=3$)	0.1050	0.4124	0.1906	0.0401	0.1583	0.0149	0.0294	0.0283
JPEG (Q=70)	0.1890	0.4536	0.0439	0.0577	0.1803	0.0172	0.0294	0.0283
JPEG (Q=90)	0.3483	0.5015	0.0502	0.0737	0.1976	0.0215	0.0294	0.0283
Noise ($\sigma=0.001$)	0.4478	0.5030	0.1300	0.0695	0.1984	0.0164	0.0294	0.0283
Noise ($\sigma=0.003$)	0.4490	0.5031	0.1298	0.0695	0.1982	0.0163	0.0294	0.0283
S&P ($p=0.001$)	0.4782	0.5056	0.1998	0.0676	0.1920	0.0149	0.0294	0.0283
S&P ($p=0.003$)	0.5178	0.5096	0.1976	0.0640	0.1788	0.0128	0.0294	0.0283

(c) CIDR

Post-processing	StarGAN	StyleCLIP	DiffAE	SimSwap	pSp-mix	BlendFace	DiffSwap	DiffFace
No Post-processing	0.1003	0.3288	0.0215	0.3709	0.1172	0.2078	0.0459	0.0535
Blur ($\sigma=1$)	0.1003	0.3288	0.0215	0.3709	0.1172	0.2078	0.0639	0.0530
Blur ($\sigma=3$)	0.0279	0.2414	0.0344	0.2113	0.0902	0.1972	0.0573	0.0477
JPEG (Q=70)	0.0084	0.2902	0.0226	0.3064	0.1060	0.2381	0.0663	0.0488
JPEG (Q=90)	0.0383	0.3285	0.0318	0.3929	0.1177	0.2848	0.0606	0.0513
Noise ($\sigma=0.001$)	0.1001	0.3287	0.0215	0.3708	0.1172	0.2078	0.0618	0.0595
Noise ($\sigma=0.003$)	0.1017	0.3287	0.0216	0.3709	0.1171	0.2055	0.0867	0.0755
S&P ($p=0.001$)	0.1379	0.3248	0.0463	0.3623	0.1129	0.1877	0.0701	0.0663
S&P ($p=0.003$)	0.1768	0.3218	0.0637	0.3540	0.1039	0.1609	0.0745	0.0795

(d) BRISQUE

Post-processing	StarGAN	StyleCLIP	DiffAE	SimSwap	pSp-mix	BlendFace	DiffSwap	DiffFace
No Post-processing	33.9631	13.3239	33.4323	1.4151	6.1890	2.2879	4.0877	1.7187
Blur ($\sigma=1$)	33.9614	13.3217	33.4494	1.4151	6.2113	2.2824	-4.8291	1.7187
Blur ($\sigma=3$)	25.4032	29.2013	4.0871	1.4561	6.7555	2.2844	-4.7393	1.7187
JPEG (Q=70)	23.7501	13.8232	13.7654	1.4813	6.4292	2.3695	-4.9117	1.7187
JPEG (Q=90)	28.2732	13.5132	14.1164	1.4518	6.2196	2.2799	-4.6799	1.7187
Noise ($\sigma=0.001$)	33.9377	13.3278	33.4539	1.4258	6.1950	2.2775	-4.9241	1.7187
Noise ($\sigma=0.003$)	34.0946	13.2977	33.4567	1.4253	6.2000	2.2636	-4.5391	1.7187
S&P ($p=0.001$)	35.2406	13.5996	35.1595	1.3816	6.2819	2.2891	-4.6808	1.7187
S&P ($p=0.003$)	37.3602	14.1166	44.4481	1.3522	6.6610	2.2468	-4.7007	1.7187

Table 15. Quantitative robustness evaluation of Anti-Forgery under various post-processing transformations.

(a) MSE

Post-processing	StarGAN	StyleCLIP	DiffAE	SimSwap	pSp-mix	BlendFace	DiffSwap	DiffFace
No Post-processing	1.6180	0.8045	0.0020	0.0190	0.2496	0.0383	0.0043	0.0065
Blur ($\sigma=1$)	1.6180	0.8045	0.0020	0.0190	0.2496	0.0383	0.0043	0.0065
Blur ($\sigma=3$)	0.0130	0.1248	0.0041	0.0007	0.0955	0.0066	0.0043	0.0065
JPEG (Q=70)	0.0016	0.0279	0.0022	0.0000	0.0253	0.0000	0.0043	0.0065
JPEG (Q=90)	0.0012	0.0681	0.0021	0.0002	0.0591	0.0001	0.0043	0.0065
Noise ($\sigma=0.001$)	1.6165	0.8015	0.0021	0.0190	0.2493	0.0377	0.0043	0.0065
Noise ($\sigma=0.003$)	1.6052	0.7892	0.0032	0.0189	0.2483	0.0325	0.0043	0.0065
S&P ($p=0.001$)	0.7597	0.6411	0.0082	0.0172	0.2191	0.0114	0.0043	0.0065
S&P ($p=0.003$)	0.4791	0.4531	0.0107	0.0145	0.1701	0.0023	0.0043	0.0065

(b) LPIPS

Post-processing	StarGAN	StyleCLIP	DiffAE	SimSwap	pSp-mix	BlendFace	DiffSwap	DiffFace
No Post-processing	0.6950	0.4493	0.0315	0.0577	0.2038	0.0412	0.0309	0.0281
Blur ($\sigma=1$)	0.6950	0.4493	0.0315	0.0577	0.2038	0.0412	0.0309	0.0281
Blur ($\sigma=3$)	0.0931	0.3330	0.0931	0.0050	0.1196	0.0137	0.0309	0.0281
JPEG (Q=70)	0.0159	0.1031	0.0175	0.0007	0.0548	0.0003	0.0309	0.0281
JPEG (Q=90)	0.0121	0.1706	0.0206	0.0017	0.0872	0.0007	0.0309	0.0281
Noise ($\sigma=0.001$)	0.6944	0.4491	0.0329	0.0577	0.2037	0.0407	0.0309	0.0281
Noise ($\sigma=0.003$)	0.6891	0.4484	0.0483	0.0575	0.2031	0.0364	0.0309	0.0281
S&P ($p=0.001$)	0.5664	0.4434	0.1905	0.0541	0.1918	0.0171	0.0309	0.0281
S&P ($p=0.003$)	0.5683	0.4462	0.2044	0.0492	0.1742	0.0053	0.0309	0.0281

(c) CIDR

Post-processing	StarGAN	StyleCLIP	DiffAE	SimSwap	pSp-mix	BlendFace	DiffSwap	DiffFace
No Post-processing	0.4103	0.2809	0.0105	0.2935	0.0781	0.3668	0.0430	0.0361
Blur ($\sigma=1$)	0.4103	0.2809	0.0105	0.2935	0.0781	0.3668	0.0502	0.0381
Blur ($\sigma=3$)	0.0402	0.1209	0.0259	0.0253	0.0346	0.2262	0.0479	0.0386
JPEG (Q=70)	0.0183	0.0512	0.0124	0.0067	0.0182	0.0080	0.0600	0.0365
JPEG (Q=90)	0.0094	0.0746	0.0113	0.0112	0.0249	0.0131	0.0537	0.0409
Noise ($\sigma=0.001$)	0.4110	0.2806	0.0115	0.2934	0.0781	0.3636	0.0512	0.0404
Noise ($\sigma=0.003$)	0.4102	0.2794	0.0181	0.2930	0.0778	0.3443	0.0819	0.0486
S&P ($p=0.001$)	0.3270	0.2689	0.0627	0.2810	0.0714	0.2325	0.0524	0.0443
S&P ($p=0.003$)	0.3124	0.2558	0.0728	0.2669	0.0631	0.0741	0.0637	0.0496

(d) BRISQUE

Post-processing	StarGAN	StyleCLIP	DiffAE	SimSwap	pSp-mix	BlendFace	DiffSwap	DiffFace
No Post-processing	29.6181	21.0689	12.5174	1.2399	6.2846	2.1080	3.5432	1.7217
Blur ($\sigma=1$)	29.6181	21.0695	12.5174	1.2399	6.2941	2.1131	-4.9171	1.7217
Blur ($\sigma=3$)	23.7931	40.2378	16.3093	1.2516	7.1598	2.4618	-4.7663	1.7217
JPEG (Q=70)	17.0328	12.7457	12.7115	1.3822	6.6151	2.5316	-5.0345	1.7217
JPEG (Q=90)	17.8007	13.8513	11.6284	1.3743	6.7212	2.4979	-4.8489	1.7217
Noise ($\sigma=0.001$)	29.5659	20.9855	11.6774	1.2426	6.3005	2.1284	-4.6537	1.7217
Noise ($\sigma=0.003$)	28.7932	20.6193	13.5594	1.2438	6.3168	2.1217	-4.2948	1.7217
S&P ($p=0.001$)	22.8764	18.2544	47.0265	1.2638	6.5509	2.3390	-4.7678	1.7217
S&P ($p=0.003$)	30.5374	15.5473	55.8376	1.2535	7.0424	2.3709	-4.8391	1.7217

Table 16. Quantitative robustness evaluation of Latent Attack under various post-processing transformations.

(a) MSE

Post-processing	StarGAN	StyleCLIP	DiffAE	SimSwap	pSp-mix	BlendFace	DiffSwap	DiffFace
No Post-processing	0.0958	0.2117	0.0619	0.0123	0.0830	0.0040	0.0097	0.0142
Blur ($\sigma=1$)	0.0958	0.2117	0.0618	0.0123	0.0830	0.0040	0.0097	0.0112
Blur ($\sigma=3$)	0.0067	0.0403	0.0178	0.0006	0.0341	0.0025	0.0097	0.0071
JPEG (Q=70)	0.0036	0.0470	0.0050	0.0007	0.0406	0.0011	0.0097	0.0073
JPEG (Q=90)	0.0157	0.0884	0.0089	0.0058	0.0607	0.0018	0.0097	0.0098
Noise ($\sigma=0.001$)	0.0958	0.2111	0.0619	0.0123	0.0830	0.0040	0.0097	0.0142
Noise ($\sigma=0.003$)	0.0958	0.2108	0.0620	0.0123	0.0829	0.0040	0.0097	0.0126
S&P ($p=0.001$)	0.0996	0.1677	0.0453	0.0119	0.0773	0.0038	0.0097	0.0140
S&P ($p=0.003$)	0.1069	0.1206	0.0211	0.0111	0.0690	0.0034	0.0097	0.0135

(b) LPIPS

Post-processing	StarGAN	StyleCLIP	DiffAE	SimSwap	pSp-mix	BlendFace	DiffSwap	DiffFace
No Post-processing	0.4368	0.6439	0.4598	0.0653	0.2118	0.0151	0.0693	0.0528
Blur ($\sigma=1$)	0.4368	0.6439	0.4598	0.0653	0.2118	0.0151	0.0693	0.0462
Blur ($\sigma=3$)	0.0977	0.3922	0.2018	0.0055	0.1154	0.0103	0.0693	0.0317
JPEG (Q=70)	0.0624	0.4322	0.0640	0.0064	0.1295	0.0051	0.0693	0.0314
JPEG (Q=90)	0.1072	0.5679	0.1151	0.0374	0.1686	0.0079	0.0693	0.0420
Noise ($\sigma=0.001$)	0.4368	0.6439	0.4597	0.0653	0.2118	0.0151	0.0693	0.0528
Noise ($\sigma=0.003$)	0.4370	0.6438	0.4593	0.0652	0.2114	0.0151	0.0693	0.0513
S&P ($p=0.001$)	0.4540	0.6382	0.4507	0.0640	0.2017	0.0146	0.0693	0.0528
S&P ($p=0.003$)	0.4786	0.6252	0.3099	0.0615	0.1872	0.0134	0.0693	0.0528

(c) CIDR

Post-processing	StarGAN	StyleCLIP	DiffAE	SimSwap	pSp-mix	BlendFace	DiffSwap	DiffFace
No Post-processing	0.1713	0.5852	0.0117	0.6140	0.2070	0.5364	0.2674	0.6911
Blur ($\sigma=1$)	0.1713	0.5852	0.0117	0.6140	0.2070	0.5364	0.2437	0.5457
Blur ($\sigma=3$)	0.0428	0.3360	0.1253	0.0337	0.1079	0.3825	0.0513	0.0611
JPEG (Q=70)	0.0266	0.3694	0.0641	0.0471	0.1250	0.2012	0.1021	0.0878
JPEG (Q=90)	0.0402	0.4821	0.1302	0.3962	0.1660	0.3213	0.2173	0.4416
Noise ($\sigma=0.001$)	0.1708	0.5852	0.0117	0.6140	0.2069	0.5364	0.2686	0.6909
Noise ($\sigma=0.003$)	0.1709	0.5851	0.0113	0.6138	0.2068	0.5362	0.2588	0.6379
S&P ($p=0.001$)	0.1753	0.5773	0.0174	0.6048	0.2007	0.5243	0.2617	0.6864
S&P ($p=0.003$)	0.1891	0.5583	0.1433	0.5881	0.1827	0.4908	0.2546	0.6645

(d) BRISQUE

Post-processing	StarGAN	StyleCLIP	DiffAE	SimSwap	pSp-mix	BlendFace	DiffSwap	DiffFace
No Post-processing	37.0453	12.2354	28.0547	0.8187	5.0783	2.3089	3.5636	1.3969
Blur ($\sigma=1$)	37.0473	12.2369	28.0543	0.8187	5.0929	2.3143	-4.9853	1.3969
Blur ($\sigma=3$)	25.8909	20.4972	19.7663	1.2985	6.5402	2.2945	-4.8327	1.3969
JPEG (Q=70)	18.3917	10.4782	18.4249	1.3436	5.5357	2.3985	-4.8356	1.3969
JPEG (Q=90)	17.9137	11.1897	22.1910	1.2787	5.0502	2.2447	-4.6161	1.3969
Noise ($\sigma=0.001$)	37.0622	12.2373	28.0319	0.8243	5.0980	2.3101	-5.2181	1.3969
Noise ($\sigma=0.003$)	37.0252	12.2361	28.1310	0.8229	5.1026	2.3068	-4.9371	1.3969
S&P ($p=0.001$)	37.3004	12.0137	40.3490	0.8119	5.2966	2.3233	-5.0029	1.3969
S&P ($p=0.003$)	37.9511	12.4351	59.4013	0.9720	5.7272	2.3136	-4.7754	1.3969

Table 17. Quantitative robustness evaluation of SCOL under various post-processing transformations.

(a) MSE

Post-processing	StarGAN	StyleCLIP	DiffAE	SimSwap	pSp-mix	BlendFace	DiffSwap	DiffFace
No Post-processing	0.0590	0.0481	0.0546	0.0050	0.0463	0.0013	0.005	0.0101
Blur ($\sigma=1$)	0.0590	0.0481	0.0528	0.0050	0.0463	0.0013	0.005	0.0095
Blur ($\sigma=3$)	0.0562	0.0490	0.0543	0.0051	0.0470	0.0013	0.005	0.0096
JPEG (Q=70)	0.0575	0.0476	0.0539	0.0050	0.0449	0.0013	0.005	0.0097
JPEG (Q=90)	0.0571	0.0481	0.0538	0.0050	0.0459	0.0013	0.005	0.0096
Noise ($\sigma=0.001$)	0.0591	0.0482	0.0526	0.0050	0.0463	0.0013	0.005	0.0097
Noise ($\sigma=0.003$)	0.0593	0.0482	0.0519	0.0050	0.0464	0.0013	0.005	0.0099
S&P ($p=0.001$)	0.0892	0.0493	0.0541	0.0052	0.0470	0.0013	0.005	0.0098
S&P ($p=0.003$)	0.1112	0.0526	0.0563	0.0054	0.0496	0.0014	0.005	0.0098

(b) LPIPS

Post-processing	StarGAN	StyleCLIP	DiffAE	SimSwap	pSp-mix	BlendFace	DiffSwap	DiffFace
No Post-processing	0.2317	0.3344	0.1759	0.0315	0.1058	0.0059	0.0309	0.0395
Blur ($\sigma=1$)	0.2317	0.3344	0.3506	0.0315	0.1058	0.0059	0.0309	0.0393
Blur ($\sigma=3$)	0.2260	0.3577	0.4508	0.0320	0.1081	0.0059	0.0309	0.0395
JPEG (Q=70)	0.2133	0.2695	0.3522	0.0312	0.1052	0.0062	0.0309	0.0390
JPEG (Q=90)	0.2162	0.3205	0.3616	0.0313	0.1055	0.0059	0.0309	0.0387
Noise ($\sigma=0.001$)	0.2322	0.3345	0.3467	0.0315	0.1058	0.0059	0.0309	0.0395
Noise ($\sigma=0.003$)	0.2369	0.3357	0.3176	0.0316	0.1059	0.0060	0.0309	0.0395
S&P ($p=0.001$)	0.4084	0.3486	0.2926	0.0325	0.1129	0.0061	0.0309	0.0395
S&P ($p=0.003$)	0.4941	0.3843	0.2882	0.0333	0.1242	0.0062	0.0309	0.0395

(c) CIDR

Post-processing	StarGAN	StyleCLIP	DiffAE	SimSwap	pSp-mix	BlendFace	DiffSwap	DiffFace
No Post-processing	0.2391	0.3939	0.3769	0.3519	0.1679	0.2293	0.1139	0.4115
Blur ($\sigma=1$)	0.2391	0.3939	0.3869	0.3519	0.1679	0.2293	0.1251	0.4022
Blur ($\sigma=3$)	0.2391	0.4076	0.4004	0.3527	0.1733	0.2345	0.1161	0.4072
JPEG (Q=70)	0.2382	0.3987	0.4054	0.3519	0.1707	0.2410	0.1315	0.4022
JPEG (Q=90)	0.2318	0.3957	0.4020	0.3505	0.1688	0.2318	0.1244	0.3993
Noise ($\sigma=0.001$)	0.2399	0.3939	0.3870	0.3520	0.1679	0.2293	0.1187	0.4068
Noise ($\sigma=0.003$)	0.2419	0.3940	0.3892	0.3521	0.1680	0.2293	0.1537	0.4104
S&P ($p=0.001$)	0.2556	0.3972	0.4201	0.3552	0.1680	0.2278	0.1325	0.4044
S&P ($p=0.003$)	0.2684	0.4081	0.4324	0.3602	0.1722	0.2326	0.1406	0.3944

(d) BRISQUE

Post-processing	StarGAN	StyleCLIP	DiffAE	SimSwap	pSp-mix	BlendFace	DiffSwap	DiffFace
No Post-processing	21.8716	6.5536	38.4324	1.7359	5.9274	2.4045	3.8448	2.0021
Blur ($\sigma=1$)	21.8667	6.5527	63.4258	1.7359	5.9304	2.4065	-4.8641	2.0021
Blur ($\sigma=3$)	27.6213	34.1819	4.2680	1.6475	6.4855	2.4577	-5.0424	2.0021
JPEG (Q=70)	22.9816	12.0078	18.6510	1.7836	6.5595	2.5256	-4.8158	2.0021
JPEG (Q=90)	20.5043	6.5675	13.8583	1.7642	5.9662	2.4475	-4.9022	2.0021
Noise ($\sigma=0.001$)	21.8907	6.5516	62.7166	1.7200	5.9241	2.4163	-4.8007	1.9668
Noise ($\sigma=0.003$)	22.1237	6.5594	58.1992	1.7190	5.8789	2.4151	-4.5728	1.8562
S&P ($p=0.001$)	26.5708	6.9881	34.9108	1.6219	6.3522	2.4385	-4.6815	2.0021
S&P ($p=0.003$)	32.1346	8.6657	67.2802	1.6968	6.8238	2.4335	-4.5129	1.9756

Table 18. Quantitative robustness evaluation of NullSwap under various post-processing transformations.

(a) MSE

Post-processing	StarGAN	StyleCLIP	DiffAE	SimSwap	pSp-mix	BlendFace	DiffSwap	DiffFace
No Post-processing	0.0054	0.0108	0.0053	0.0060	0.0044	0.0018	0.0053	0.0103
Blur ($\sigma=1$)	0.0054	0.0108	0.0154	0.0060	0.0044	0.0018	0.0053	0.0100
Blur ($\sigma=3$)	0.0060	0.0156	0.0165	0.0061	0.0071	0.0018	0.0053	0.0103
JPEG (Q=70)	0.0046	0.0114	0.0165	0.0060	0.0046	0.0018	0.0053	0.0103
JPEG (Q=90)	0.0042	0.0108	0.0163	0.0060	0.0044	0.0018	0.0053	0.0103
Noise ($\sigma=0.001$)	0.0055	0.0108	0.0154	0.0060	0.0044	0.0017	0.0053	0.0103
Noise ($\sigma=0.003$)	0.0062	0.0108	0.0153	0.0060	0.0044	0.0017	0.0053	0.0103
S&P ($p=0.001$)	0.0526	0.0124	0.0175	0.0061	0.0069	0.0017	0.0053	0.0103
S&P ($p=0.003$)	0.0816	0.0161	0.0202	0.0061	0.0101	0.0018	0.0053	0.0103

(b) LPIPS

Post-processing	StarGAN	StyleCLIP	DiffAE	SimSwap	pSp-mix	BlendFace	DiffSwap	DiffFace
No Post-processing	0.0340	0.1046	0.0353	0.0358	0.0254	0.0080	0.0316	0.0410
Blur ($\sigma=1$)	0.0340	0.1046	0.2684	0.0358	0.0254	0.0080	0.0316	0.0403
Blur ($\sigma=3$)	0.0647	0.2558	0.2753	0.0364	0.0352	0.0083	0.0316	0.0410
JPEG (Q=70)	0.0314	0.1144	0.2388	0.0358	0.0261	0.0080	0.0316	0.0410
JPEG (Q=90)	0.0284	0.1049	0.2511	0.0358	0.0255	0.0080	0.0316	0.0410
Noise ($\sigma=0.001$)	0.0347	0.1047	0.2690	0.0358	0.0254	0.0080	0.0316	0.0410
Noise ($\sigma=0.003$)	0.0424	0.1051	0.2686	0.0358	0.0254	0.0077	0.0316	0.0410
S&P ($p=0.001$)	0.2871	0.1286	0.2133	0.0361	0.0410	0.0078	0.0316	0.0410
S&P ($p=0.003$)	0.3953	0.2088	0.1904	0.0366	0.0575	0.0079	0.0316	0.0410

(c) CIDR

Post-processing	StarGAN	StyleCLIP	DiffAE	SimSwap	pSp-mix	BlendFace	DiffSwap	DiffFace
No Post-processing	0.2792	0.5677	0.5028	0.5114	0.2856	0.3983	0.2131	0.5627
Blur ($\sigma=1$)	0.2792	0.5677	0.4739	0.5114	0.2856	0.3983	0.2076	0.5659
Blur ($\sigma=3$)	0.2811	0.5726	0.4748	0.5120	0.2909	0.3976	0.2065	0.5671
JPEG (Q=70)	0.2807	0.5684	0.4755	0.5116	0.2864	0.3990	0.2079	0.5661
JPEG (Q=90)	0.2764	0.5678	0.4746	0.5112	0.2857	0.3986	0.2128	0.5610
Noise ($\sigma=0.001$)	0.2796	0.5676	0.4741	0.5114	0.2856	0.3989	0.2049	0.5666
Noise ($\sigma=0.003$)	0.2811	0.5675	0.4739	0.5115	0.2856	0.3989	0.2264	0.5659
S&P ($p=0.001$)	0.2910	0.5680	0.4865	0.5122	0.2856	0.3987	0.2119	0.5660
S&P ($p=0.003$)	0.2951	0.5696	0.4937	0.5100	0.2871	0.4015	0.1965	0.5703

(d) BRISQUE

Post-processing	StarGAN	StyleCLIP	DiffAE	SimSwap	pSp-mix	BlendFace	DiffSwap	DiffFace
No Post-processing	18.3381	12.8330	28.2317	1.0041	5.6176	2.4165	3.7283	1.8001
Blur ($\sigma=1$)	18.3371	12.8382	23.4864	1.0041	5.6122	2.4151	-4.6729	1.8001
Blur ($\sigma=3$)	25.8780	39.9484	14.1503	1.0441	5.7040	2.4535	-5.1286	1.7832
JPEG (Q=70)	17.1786	10.0821	12.9199	0.9735	5.5468	2.4049	-4.7219	1.8001
JPEG (Q=90)	18.4471	12.9163	10.1845	1.0331	5.6290	2.4007	-5.0886	1.6849
Noise ($\sigma=0.001$)	18.4011	12.7664	24.1845	0.9946	5.6112	2.4029	-4.9700	1.6655
Noise ($\sigma=0.003$)	18.7338	12.3520	28.1653	0.9987	5.6128	2.3489	-4.4995	1.7538
S&P ($p=0.001$)	23.5852	10.2667	44.5269	0.9955	6.0849	2.3170	-5.0703	1.8001
S&P ($p=0.003$)	30.7863	8.8471	57.5830	1.0936	6.4697	2.2831	-4.9616	1.8001

D. Transferability

This appendix provides comprehensive transferability results across all evaluated generators and defense methods. While Figure 4 visualizes heatmaps based on MSE and CIDR, we additionally report results for other evaluation metrics to enable a more complete and transparent analysis.

Source Generators denote the generator used to generate perturbations, and Target Generators denote the generator used for evaluation. In this section, we consider a black-box transfer setting, where the source and target generators are different. Specifically, perturbations generated on a given source generator are applied to different target generators, and their effects are evaluated by measuring discrepancies between the generated outputs $G(x_{adv})$ and the corresponding clean outputs $G(x)$. All transferability experiments are conducted without access to the architecture or parameters of the target generator.

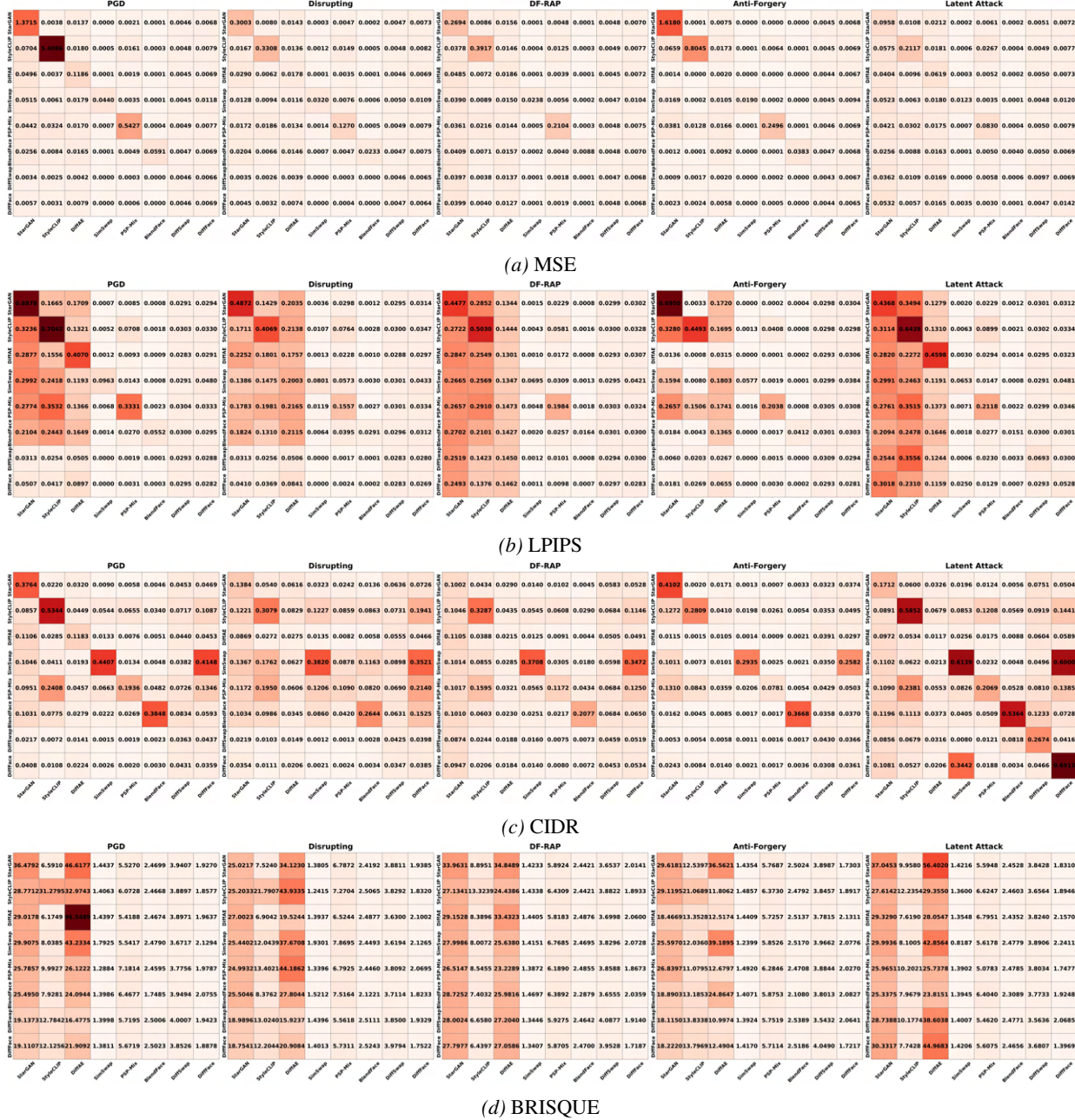


Figure 12. Transferability of proactive defenses across deepfake generators.

E. Correlation between Evaluation Metrics

We analyze the pairwise Spearman correlations among representative metrics across various generators and defenses, as summarized in Figure 13. First, we observe strong redundancy between pixel-level metrics (MSE and PSNR). However, the relationship between pixel-level and perceptual fidelity varies by generator architecture. Face swap models exhibit consistent degradation patterns across all fidelity metrics, whereas attribute manipulation models show high variance, indicating that defenses can degrade pixel-level fidelity and perceptual quality independently.

Second, identity-based metrics generally demonstrate weaker correlations with synthesis fidelity than the inter-correlation among fidelity metrics themselves. This observation necessitates the simultaneous use of both metric types for a holistic evaluation. Notably, CIDR exhibits a stronger correlation with ID Loss than with synthesis metrics, maintaining a low correlation with synthesis metrics. This independence suggests that CIDR successfully isolates defense-induced identity disruption by mitigating the generator-specific biases inherent in ID Loss. Finally, BRISQUE shows minimal correlation with all other metrics, capturing an orthogonal dimension of image degradation.

MSE	1.0000	0.9608 ± 0.123	0.4732 ± 0.199	0.4227 ± 0.239	0.1344 ± 0.078	0.2794 ± 0.202	0.2268 ± 0.187
PSNR	0.9608 ± 0.123	1.0000	0.4692 ± 0.216	0.4414 ± 0.221	0.1367 ± 0.080	0.2618 ± 0.190	0.1997 ± 0.144
SSIM	0.4732 ± 0.199	0.4692 ± 0.216	1.0000	0.5480 ± 0.227	0.0967 ± 0.060	0.2504 ± 0.187	0.2421 ± 0.162
LPIPS	0.4227 ± 0.239	0.4414 ± 0.221	0.5480 ± 0.227	1.0000	0.0873 ± 0.082	0.3797 ± 0.181	0.1927 ± 0.150
CIDR	0.1344 ± 0.078	0.1367 ± 0.080	0.0967 ± 0.060	0.0873 ± 0.082	1.0000	0.2016 ± 0.176	0.0699 ± 0.060
ID Loss	0.2794 ± 0.202	0.2618 ± 0.190	0.2504 ± 0.187	0.3797 ± 0.181	0.2016 ± 0.176	1.0000	0.1541 ± 0.116
BRISQUE	0.2268 ± 0.187	0.1997 ± 0.144	0.2421 ± 0.162	0.1927 ± 0.150	0.0699 ± 0.060	0.1541 ± 0.116	1.0000
	MSE	PSNR	SSIM	LPIPS	CIDR	ID Loss	BRISQUE

(a) Attribute Manipulation

MSE	1.0000	1.0000 ± 0.000	0.8103 ± 0.127	0.8360 ± 0.065	0.0888 ± 0.073	0.3952 ± 0.168	0.1784 ± 0.159
PSNR	1.0000 ± 0.000	1.0000	0.8103 ± 0.127	0.8360 ± 0.065	0.0888 ± 0.073	0.3952 ± 0.168	0.1784 ± 0.159
SSIM	0.8103 ± 0.127	0.8103 ± 0.127	1.0000	0.8820 ± 0.046	0.0998 ± 0.101	0.5602 ± 0.170	0.1604 ± 0.115
LPIPS	0.8360 ± 0.065	0.8360 ± 0.065	0.8820 ± 0.046	1.0000	0.0919 ± 0.082	0.4715 ± 0.136	0.1400 ± 0.114
CIDR	0.0888 ± 0.073	0.0888 ± 0.073	0.0998 ± 0.101	0.0919 ± 0.082	1.0000	0.1226 ± 0.147	0.0971 ± 0.056
ID Loss	0.3952 ± 0.168	0.3952 ± 0.168	0.5602 ± 0.170	0.4715 ± 0.136	0.1226 ± 0.147	1.0000	0.1212 ± 0.074
BRISQUE	0.1784 ± 0.159	0.1784 ± 0.159	0.1604 ± 0.115	0.1400 ± 0.114	0.0971 ± 0.056	0.1212 ± 0.074	1.0000
	MSE	PSNR	SSIM	LPIPS	CIDR	ID Loss	BRISQUE

(b) Face Swap

Figure 13. Pairwise correlation patterns of evaluation metrics across proactive defenses. Rows and columns denote evaluation metrics, with cell values representing the average absolute Spearman correlation ($\pm$ standard deviation) across all defense methods. Diagonal elements are 1.0 (self-correlation). (a) shows attribute manipulation models, and (b) shows face swap models. Darker colors indicate stronger correlations.